FrutaMIDI: capacitive USB MIDI controller

Engineering report

FrutaMIDI engineering project

FrutaMIDI v1.3 (change request CR-003), interfaces frozen-8 (frozen-9 proposed), requirements draft-14, 2026-11-11

Branch `change/CR-003` at `f6693c9`; v1.3 is not released. Previous release: V1.2 (git tag `v1.2.0`)

Abstract

FrutaMIDI is a USB MIDI controller that sends notes when a user touches conductive objects wired to a Raspberry Pi Pico and an Adafruit MPR121 capacitive sensor module, built on a breadboard or on a single-sided home-etched PCB. This report collects the requirements, decisions, interfaces, analyses and host tests for FrutaMIDI. **Version 1.3** (change request CR-003, Section 3) converts the PCB to surface-mount parts on one copper layer on the component side, for home chemical etching without drilling: 4 mounting holes remain instead of 133. The electrode resistors become standard 2512 chip resistors whose survival of a 4 kV discharge is not proven, a choice the user accepted to keep the core BOM at 14.84–16.84 EUR excluding VAT and shipping (CALCULATION). An independent review (T-102) found the first layout not ready for a first build; its fifteen findings were fixed or ruled (ADR-028). The sections after Section 3 keep the v1.2 analyses for traceability. Version 1.2 (change request CR-002) replaced the electrode series resistors by pulse-rated parts lying flat. By analysis of the design files, a 4 kV contact discharge should then neither arc across a resistor nor exceed its pulse rating; this has not been tested. A manager PCB design review then widened the JP1 solder-jumper gaps, enlarged the screw-terminal pads, tied the potentiometer tabs to AGND and added a 100 nF wiper filter (ADR-021). A final independent re-review of these changes (T-079) found no critical or major issue (verdict release-ready); its seven findings were fixed or accepted (ADR-022), among them a 0.9 mm drill for C1 and a plastic knob on an insulated shaft for the potentiometer (risk R-011). Section 13 lists the changes since the V1_BASELINE release. Every number is copied from a cited project file with its label. **No part of the hardware has been built or measured**: analyses, simulations, design-rule checks and host tests are not hardware validation, and the physical validation plan in Section 11 lists what remains to be measured.

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1 Introduction and scope

FrutaMIDI is a USB MIDI controller that turns touches on conductive objects (fruit, plants, water) into MIDI notes. It uses a Raspberry Pi Pico board and an Adafruit MPR121 breakout (product 1982). The project continues an existing repository (v1.0.0 released) and covers two builds: a solderless breadboard and a single-sided PCB for home etching, now at label "FrutaMIDI v1.3" [`management/DECISIONS.md`] (ADR-000; ADR-023). Version 1.3 is the result of change request CR-003 [`changes/CR-003.md`], which converts the board to surface-mount parts for chemical etching at home; it is described in Section 3. Version 1.2 (CR-002, [`changes/CR-002.md`]) was the through-hole board. The changes since the V1_BASELINE release are listed in Section 13.

How to read this report. Section 3 describes the current v1.3 board: its geometry, BOM and cost, ESD analysis, review findings, first-build measurements and requirement status. Sections 4 to 11 were written for the v1.2 through-hole board and are kept for traceability; the firmware, host software and pin map in them still apply (REQ-031). Where they disagree with Section 3 on the board, its parts or its numbers, Section 3 applies.

1.1 Scope

- In scope: electronics design (KiCad, generated by scripts), Pico firmware, the browser sampler (`docs/index.html`), host tests, user documentation and this report [`CLAUDE.md`].
- Out of scope: ordering parts, flashing or powering hardware, and any physical measurement. Hence every [PHYSICAL] requirement is at most PARTIAL or NOT RUN [`management/REQUIREMENTS.md`].
- Hosts: Windows 10, Windows 11 and macOS 12 or later (ADR-007 D-6).

1.2 Document conventions

- Every number carries one label: SOURCE, CALCULATION, SIMULATION, MEASUREMENT or ASSUMPTION, and a citation in brackets to the repository file it is copied from.
- Status is reported only as PASS, PARTIAL, FAIL, BLOCKED or NOT RUN.
- Passing ERC, DRC or a compile is not hardware validation.

2 Requirements and verification status

Source: [`management/REQUIREMENTS.md`] and [`requirements/requirements.csv`], revision draft-14 (T-106, 2026-11-11, after documentation T-105; draft-13, T-104, covered the T-103 board after the CR-003 review, ADR-028), proposed and not yet baselined. Drafts 10 to 14 rewrote REQ-027, REQ-029 and REQ-030 for the SMD board (ADR-023) and updated the evidence of REQ-006, REQ-023, REQ-025, REQ-028, REQ-032, REQ-036 and REQ-038. Status changes during CR-003: REQ-036 PASS to FAIL (draft-10, documents still described V1.2) and back to PASS (draft-14, interface_check 121/121 after T-105); REQ-032 FAIL to PARTIAL (draft-11) to PASS (draft-12, on the basis ruled by ADR-027 D-1). Methods: A analysis, S simulation, I inspection, T test. By the project rule, a requirement whose analysis or inspection passes while its physical test is NOT RUN is PARTIAL (ADR-009). Nothing was raised to PASS on hardware grounds.

Summary. Of 42 requirements: PASS 18, PARTIAL 18, FAIL 2, NOT RUN 4 [`requirements/requirements.csv`].

- FAIL: REQ-018 VBUS range (the USB low-power-port minimum of 4.40 V is below the Pico's recommended 4.50 V, SOURCE S-01, S-10); REQ-040 JP1 wrong-position tolerance, kept FAIL with accepted residual risk R-007 (ADR-027 D-5).

- NOT RUN (physical only): REQ-010 status LED, REQ-011 USB MIDI class compliance, REQ-037 sensing range, REQ-038 false triggers and crosstalk.
- The CR-003 rows are discussed in Section 3.

The table below is generated from `requirements/requirements.csv` by `reports/report/scripts/csv_to_table.py`; do not edit it by hand.

Table 1: Requirements and verification status (generated from requirements/requirements.csv).

ID	Requirement	Verification	Status	Evidence
REQ-001	**Note On/Off per electrode** Each of the 12 electrode inputs E0–E11 (MPR121 ELE0–ELE11) shall send one MIDI Note On, on the configured channel (default 1) with the configured fixed velocity (default 100, range 1–127), when a touch is detected. It shall send a Note Off for the same note when the touch is released.	I + T [PHYSICAL]	PARTIAL	I: PASS. The chain terminal → series resistor → module electrode is 1:1 (interface_ check `pcb.A2.pin2..13`, `pcb.J2.*`). T: NOT RUN.
REQ-002	**Note Off integrity** The Note Off for a pad shall carry the note number sent in its Note On, even if the scale or octave changed in between. All sounding notes shall be released before a recalibration.	T (host unit test) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware test NOT RUN.
REQ-003	**Second sensor (24 pads)** The firmware shall support a second MPR121 at I2C address 0x5B (`NUM_SENSORS = 2`), mapping its electrodes to pads 12–23. This is a breadboard-only option; the PCB carries one module (ADR-007 D-3).	A + T [PHYSICAL]	PARTIAL	A: latency PASS at 400 kHz (12.35 ms $\leq$ 15 ms, jitter 8.35 ms), using the worst-case run-based burst plan (budgets.out §b). T: NOT RUN.
REQ-004	**Polyphonic aftertouch** While a pad is touched, the device shall send Polyphonic Key Pressure (0xAn) with a value 0–127. The value shall map linearly from delta = baseline − filtered, clamped to [12, 120]. It shall be evaluated every 15 ms and sent only when it changes by at least 2.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Aftertouch messages are sent after all Note On/Off of the same loop (ADR-010). Hardware NOT RUN.
REQ-005	**Six scales** The device shall provide 6 scales, selected cyclically: Drum Rack (36–47, octave ignored, the power-on default), major pentatonic, minor pentatonic, major, natural minor and chromatic (root 48). Notes shall be clamped to 0–127.	A + T (host)	PASS	A: notes 24–99 with 12 pads and 24–127 with 24 pads, never clamped (T-009 budgets §d). Host tests PASS (T-040 `test_scales.py`).

ID	Requirement	Verification	Status	Evidence
REQ-006	**Octave potentiometer** The octave potentiometer (off-board on 3 wires, ADR-023 D-2) shall shift melodic scales by −2..+2 octaves using 5 ADC zones, each V33/5 = 0.660 V wide. The octave shall change only when the reading is more than 0.7 zone from the current zone centre, i.e. 0.2 zone (0.132 V) beyond the border, which is the dead band. The combined ADC noise, DNL, quantisation and supply ripple shall stay below the dead band. A static ADC offset shifts every threshold together and does not consume the dead band. The pot shall have no effect on Drum Rack and shall be enabled by default on the PCB build.	A + T (host) + T [PHYSICAL]	PARTIAL	A: PASS. Error budget 0.0364 V < dead band 0.132 V, margin 0.096 V (budgets A5). Inputs: noise from ENOB 8.7 (SOURCE S-17 Table 627); DNL peak 10 LSB12 (ASSUMPTION; S-17 §4.9.4 and erratum RP2040-E11 give peaks at codes 512/1536/2560/3584, height only in a figure); 10-bit step; ripple 10 mV (ASSUMPTION). The nearest DNL-peak code is 0.049 V from a switch point. The static offset of about 30 mV (S-01 §4.3) is reported, not budgeted (review F-003). Host tests PASS (T-043). RV1 is the Bourns PTV09A-4015F-B103, 10 kΩ ±20 % linear, insulated flatted shaft (SOURCE T96-S2/T96-S3, ADR-024/T-096), with knob SPK-016A (bore fit NOT CONFIRMED). The EE check calc_pot_rc gives ADC loading ≤ 14.9 counts PASS, residual resistance 6.25 % PASS and settling 2.89 ms (CALCULATION, T-096). *(continued below)*
	REQ-006 evidence, continued (1/1): C1 wiper filter (ADR-021, ADR-023): 100 nF X7R 1206, Samsung CL31B104KBCNNNC (187-CL31B104KBCNNNC) from POT to AGND; worst Thevenin source 3.0 kOhm (10 kOhm +20 %, SOURCE T96-S3, mid travel) and worst C 126.5 nF (+10 % tolerance, +15 % X7R temperature; ASSUMPTION, the Samsung datasheet was not opened) give tau 0.380 ms and settling to 1/2 LSB at 10 bits in 2.89 ms against the 30 ms read period (SOURCE fm_pot.h POT_READ_MS), margin 10.4x (CALCULATION, budgets A7 PASS, criterion 5x ASSUMPTION). Off-board wiring (ADR-023 D-2), budgets A8 PASS: 0.50 m at 50 pF/m adds 25 pF to C1 (0.028 % of 90 nF), and the IR drop is 26.8 uV per wire at 0.41 mA and 0.13 Ohm/m against the 0.132 V dead band (CALCULATION; length, capacitance and resistance per metre are ASSUMPTIONS). Hum and interference pickup on the wires is NOT RUN (not analysed); risk R-011 covers ESD on the wires. C1 does not enter the error budget, which keeps the 10 mV ripple ASSUMPTION. Hardware NOT RUN.			
REQ-007	**Button short press** A short press of the button (released within 800 ms, sampled every 10 ms) shall select the next scale, and the LED shall blink N times for scale N.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Hardware NOT RUN.
REQ-008	**Button long press recalibration** A press of more than 800 ms shall release all notes and recalibrate all sensors; the LED then blinks 3 times.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS (T-044 handoff, 295 host tests passed). Recalibration blocks for 16.8–22.1 ms per sensor at 400 kHz (CALCULATION); this is user-requested and allowed by REQ-016. Hardware NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-009	**Sensor absent or lost** If no MPR121 answers at start, the device shall send no MIDI from it and retry. During play, a sensor shall be marked lost after 3 consecutive failed I2C transactions, or at once when its ECR read-back (every 250 ms) differs from the value read after start-up (a chip that reset itself). On loss the firmware shall release that sensor's notes, show the fast error blink and stop reading it. It shall retry start-up when more than 2 s have passed since the last start attempt (this also covers a second sensor missing at boot). The other sensor shall keep playing.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (15 tests), within `295 passed` (T-044 handoff). Implementation: `fm_notes.h`/ health logic, `startSensor()` = begin → setClock(400000) → configure → ECR read. A self-reset chip releases its notes through the normal touch path before the ECR check, so only one Note Off is sent. Physical NOT RUN (it needs the sensor unplugged or reset during play).
REQ-010	**Status LED** The LED shall give 2 slow blinks at start when a sensor is found, and shall be lit while any pad is touched. On Pico W and Pico 2 W the LED is driven through the CYW43 (WL_GPIO0).	T [PHYSICAL]	NOT RUN	
REQ-011	**USB MIDI class compliance** The device shall enumerate as a USB MIDI 1.0 class device (MIDIStreaming, bulk endpoints). It shall work without vendor drivers on Windows 10, Windows 11 and macOS 12 or later.	T [PHYSICAL]	NOT RUN	
REQ-012	**Device name** The USB product string and the MIDI port name shall be "FrutaMIDI".	I + T [PHYSICAL]	PARTIAL	I: `DEVICE_NAME = "FrutaMIDI"` is passed to `setProductDescriptor` and `setStringDescriptor`. T: NOT RUN.
REQ-013	**MIDI message set** The device shall send only Note On (0x9n, CIN 0x9), Note Off (0x8n, velocity 0, CIN 0x8) and Polyphonic Key Pressure (0xAn, CIN 0xA), on one channel n = MIDI_CHANNEL − 1. Incoming MIDI shall be read and discarded.	I	PASS	`fm_midi.h` constants match S-14 Table 4-1.
REQ-014	**Touch-to-MIDI latency** Device latency from the first touch after a pad has been idle for at least 15 ms (DEBOUNCE_MS) to the USB-MIDI packet being available to the host shall be $\leq$ 15 ms worst case, for electrode capacitance $\leq$ 100 pF, with mitigation M1 (one burst read per sensor per aftertouch pass). A re-touch of the same pad within the 15 ms lockout is outside this requirement and is reported separately.	A + T [PHYSICAL]	PARTIAL	A: PASS. 1 sensor 10.72 ms, 2 sensors 12.35 ms at 400 kHz, including the T-044 ECR poll ($\approx$ 0.14 ms per sensor) (CALCULATION, budgets.out §b). Separate term: a re-touch inside the lockout adds up to 15 ms, giving 25.72 ms worst case (review F-011). Design limit C $\leq$ 210 pF. T: NOT RUN (plan item P-2). Flagged for user review (ADR-007).
REQ-015	**Latency jitter** Latency jitter (worst minus best) shall be $\leq$ 10 ms for electrode capacitance $\leq$ 100 pF.	A + T [PHYSICAL]	PARTIAL	A: PASS. 6.72 ms (1 sensor) and 8.35 ms (2 sensors) at 400 kHz, including the ECR poll (CALCULATION). T: NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-016	**No blocking during play** During play, no firmware action other than a user-requested recalibration or the automatic restart of a lost sensor shall block touch processing for more than 10 ms.	I + A	PASS	I: only delay(10) in midiBegin() (setup); non-blocking LED (fm_ blink.h). A: worst loop ~1.72 ms, 1 sensor, 400 kHz (sim/ systems/budgets.out, CALCULATION). Exception: lost-sensor restart 16.8-22.1 ms per retry, >=2 s apart (review N-001, ADR-018).
REQ-017	**USB bus power only** The device shall be powered only from USB VBUS through the Pico's micro-USB connector. No other power input shall exist.	I	PASS	Pico pins 39 and 40 are unconnected (`pcb.A1.no_ other_nets`).
REQ-018	**VBUS range** The device shall operate with VBUS from 4.40 V to 5.50 V at its connector.	A + T [PHYSI-CAL]	FAIL	A: VSYS 3.95–5.25 V, inside the 1.8–5.5 V range (S-01 §2.1): PASS. However, the 4.40 V low-power-port minimum (S-10) is below the Pico's recommended VBUS of 5 V ± 10 % = 4.50 V (S-01 §2.3): budgets check A3b FAIL, so the requirement as stated FAILs (review F-010). Electrically the Pico still runs (VSYS margin 2.15 V). Options: restrict the statement to 4.50–5.50 V, or accept as a documented deviation. T: NOT RUN.
REQ-019	**VBUS current** VBUS current shall be ≤ 100 mA (the declared bMaxPower, S-11) for every supported Pico board.	A + T [PHYSI-CAL]	PARTIAL	A: worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W. Lower bounds are SOURCE (S-01 Table 3, S-16 Table 2); RP2040 pull-ups 50–80 kΩ are SOURCE S-17. The upper bounds are ASSUMPTIONs. T: NOT RUN (plan item P-3).
REQ-020	**3V3 rail load** The external load on the Pico 3V3(OUT) shall be ≤ 300 mA with at least 50 % margin.	A	PASS	10.15 mA worst case with 2 sensors and PCB pull-ups: 96.6 % margin.
REQ-021	**MPR121 supply voltage** The MPR121 VDD, fed from the Pico 3V3 through the module LDO, shall be within 2.0–3.6 V.	A	PASS	3.10–3.40 V (CALCULATION; dropout and rail tolerance are ASSUMPTIONs).
REQ-022	**Touchable voltage (SELV)** Every user-touchable conductor shall be at ≤ 3.6 V relative to GND and shall be supplied only from the USB SELV source.	A	PASS	Electrodes are charged from MPR121 VDD ≤ 3.6 V (S-04), and REQ-017 holds.

ID	Requirement	Verification	Status	Evidence
REQ-023	**Electrode ESD robustness** Each user-touched electrode terminal shall survive ±4 kV contact discharge without permanent damage. The ±4 kV target is an ASSUMPTION, following IEC 61000-4-2 level 2 (the standard was not read).	I + A + T [PHYSICAL]	PARTIAL	Estimate on ASSUMPTIONs, not a PASS (review F-002). Pulse capability of the fitted part is NOT PROVEN and the user accepted that (ADR-026), so this requirement stays PARTIAL at best until the test. v1.3 design (ADR-023 D-1, ADR-026; sim/electronics/calc/check_gap.out, calc_pulse_rated.out, calc_gnd_bounce.out, sim_electrode.out). Part: R1–R12 are YAGEO RC2512FK-073K3L (3k3 ±1 %, 1 W, 2512; Mouser 603-RC2512FK-073K3L, SOURCE T96-S0b) on footprint FrutaMIDI:R_2512_HomeEtch_G5.0mm (T-098 rename, land unchanged; interface_check `pcb.R1_R12_*` and pcb_rules_check `ADR-026 R1-R12 on footprint` PASS, mutation-tested T-099). Current: 4 kV / (3.3 kΩ + 330 Ω IEC source) = 1.10 A, 1.11 A at −1 % (3267 Ω), against the 1.33 A pin limit inferred from the HBM ±2 kV rating (the limit is itself an ASSUMPTION), margin 1.20–1.22 (CALCULATION); 2.20 A at 8 kV (FAIL). *(continued below)*

REQ-023 evidence, continued (1/3): The margin compares peak current only: the 4 kV strike carries 600 nC against 200 nC for HBM 2 kV (3.0×) and decays with τ 0.54 µs against 0.15 µs (3.6×) (CALCULATION, review F3-005; stated beside the margin in hardware/README.md and reports/electronics/analysis.md section 8.3), so the peak-current margin understates the stress and is not a proof. Sensitivity with 3k3 ±1 %: worst ratio 0.904 at 3267/3300/3333 Ω against the 0.90 limit (SIMULATION, sim_electrode, PASS). Voltage across R: 3636 V at 4 kV, 7273 V at 8 kV; energy 1.09 / 4.36 mJ, τ 0.54 µs (CALCULATION). The YAGEO data give no pulse rating: 3.6 kV is 7.3× the 500 V overload voltage and 18× the 200 V working voltage (SOURCE YAGEO RC_L V.11, T96-S0); the old 182 mJ energy criterion is 167× one strike but is not a proof (calc_pulse_rated: `RC2512FK-073K3L pulse capability at IEC 4 kV: NOT PROVEN`). Flashover around R (gap rule ≥ 3.5 mm at 4 kV, ≥ 7.5 mm at 8 kV; CALCULATION on ASSUMPTION models; IEC 60664-1 F.2 recalled, not opened): (a) copper ELEx–Ex 5.00 mm and (b) Ex to other MPR121/Pico nets 5.29 mm (MEASUREMENT on the design file; land gap also pcb_rules_check `REQ-023 R1-R12 land gap` PASS 5.0 mm, 12 parts); (d) chip terminations (L 6.35 ±0.10, I1 0.60 ±0.20 mm SOURCE YAGEO RC_L V.11; placement error ASSUMPTION): worst 4.65 / 4.54 / 4.08 mm at 0 / ±0.25 / ±0.5 mm (CALCULATION); (e) mounting-metal bypass 8.40 mm (CALCULATION, T-092 holes). All PASS at 4 kV; (a), (b) and (d) FAIL at 8 kV. Ex to GND edge track 1.40 mm at every J2 pad (MEASUREMENT, pcb_rules_check `ADR-020` PASS, mutation-tested): the intentional spark gap (ADR-020), which arcs before the 5 mm gap around R.

ID	Requirement	Verification	Status	Evidence
	REQ-023 evidence, continued (2/3): Module GND to Pico GND bounce up to 330 V for about 1 ns, ≤ 1.49 A and ≤ 0.90 nC per I^2C line through R15/R16 (CALCULATION on ASSUMED inductances, T-103: taller module pin path 5.0–8.5 nH; was 236 V); 0.56× the peak and 0.2 % of the charge of the RP2040 HBM 4 kV FT-pin equivalent (budgets D1 PASS on that criterion). RP2040 HBM 4 kV rating stated for digital FT pins only; no rating found for GP26/GP27 (NOT CONFIRMED, EE-S28): risk R-010. 8 kV: risk R-006, scope stays ±4 kV. Plan B (T-097, not adopted): 12 × Bourns CDSOD323-T05LC (652-CDSOD323-T05LC, 1.0 pF typ, 5 µA) to GND with R1–R12 at 1 kΩ; sensitivity 0.938 PASS at 1 kΩ, 0.897 FAIL at 3k3 (SIMULATION); +6.18 EUR; before adoption confirm the maximum C_d and the leakage at 3.3 V. Trigger: risk R-018, accepted by the manager (ADR-027 D-2). RV1 Bourns PTV09A-4015F-B103 (off-board, insulated flatted shaft, SOURCE T96-S2/T96-S3) with knob SPK-016A (bore fit NOT CONFIRMED): the V1.2 hard copper link from the frame to AGND is gone; a metal shaft would not be protected (internal gap 0.5–2 mm ASSUMPTION, below the 3.5 mm rule; no series R on the three wires; C1 holds a wiper strike to about 6 V at 4 kV, CALCULATION, order of magnitude, calc_r011_offboard_pot.out, does not prove the ADC pad survives): controlled by the insulated shaft and insulating knob (hardware/README.md) and risk R-011. T: NOT RUN [PHYSICAL]. Plan: (1) before the test, measure the resistance of every R1–R12 (expected 3267–3333 Ω) and the body length L and termination length I1 of one part with a caliper; inside the analysed range if 6.25 ≤ L ≤ 6.45 mm and 0.40 ≤ I1 ≤ 0.80 mm (SOURCE YAGEO RC_L V.11), otherwise re-run check_gap (d); clean flux from around R1–R12; *REQ-023 evidence, continued (3/3):* (2) ESD gun, ±4 kV contact on each pad E0–E11 of J2 with the device enumerated and playing; after each strike record whether the arc goes to the GND edge track or across a resistor, any I^2C error, sensor-lost blink (REQ-009), USB disconnect or reset; (2b) strike the RV1 knob and, with the knob removed, the shaft tip, ±4 kV contact (R-011), recording the same effects plus the pot reading before and after; if the fitted shaft or knob is not the specified insulating type, record it and treat the result as not representative; record the wire length to RV1 and any metal enclosure; (3) after the series, measure every R1–R12 again; pass = every pad still detects touch and sends MIDI, RV1 still sweeps the full CC range, and every R1–R12 stays within 3234–3366 Ω (±2 %, ASSUMPTION from the 0.901 sensitivity at 3366 Ω, SIMULATION T-094); temporary upsets that recover without a power cycle are recorded, not a fail. A failure of (3), or an arc across a resistor in (2), is the plan-B trigger (R-018, ADR-027 D-2).			
REQ-024	**I2C hand-wire fault current** A misplaced SDA or SCL hand wire (including one on a supply pad S1/S2/S6/S7) shall not drive an RP2040 pad beyond its IO current limits.	I + A	PASS	I: 220 Ω series resistors are present between the Pico pads and the bus: pad 31 SDA_PICO → R16 → SDA, pad 32 SCL_PICO → R15 → SCL (interface_check `fw.pcb.PIN_SDA/SCL_vs_pcb`). A: fault current 15.5 mA per line at 3.4 V, 30.9 mA for both lines, ≤ IIOVDD_MAX 50 mA (SOURCE S-17 Table 625). Low-level margins: Pico→MPR121 0.138 V, MPR121→Pico 0.235 V (budgets B5/B6, CALCULATION). Statement reworded after ADR-009 D-9: a wire on a supply pad is still physically possible, but it is now current-limited.

ID	Requirement	Verification	Status	Evidence
REQ-025	**Both module revisions** One PCB shall accept both Adafruit 1982 revisions. JP1, a 3-pad surface solder bridge, connects ORIG (+3V3 → S1 = VIN) or QT (+3V3 → S7 = VCC). The SDA/SCL wires from the J1 surface pads go to S5/S6 (original) or S3/S4 (STEMMA QT).	I + T [PHYSICAL]	PARTIAL	I: PASS on the design file (T-093). interface_check `pcb.JP1_selector`, `pcb.JP1_smd_bridge`, `pcb.J1_J3_smd` (SMD pads, no holes), `pcb.J1` (J1 1 = SCL, 2 = SDA). hardware/README.md section 5 table (JP1, SDA, SCL) = ORIG/S5/S6 and QT/S3/S4, which matches the Eagle-derived S-row (`doc.hw_readme.jp1_and_wires.*`). pcb_rules_check: JP1 pad gap 1.87 mm against the 1.8 mm rule, mutation-tested. Change from V1.2: the board has no silkscreen (ADR-023), so ORIG and QT are etched in copper and the longer legend 'ORIG = no white connectors', 'QT = 2 white connectors' exists only on the assembly map (`pcb.JP1_legend`, F.Fab); see REQ-040 and risk R-007. T-103: A2 lands now 1.8 × 4.2 mm (electrode row) and 1.8 × 5.515 mm (S row, joined to its J3 pad), 1.73–1.775 mm beyond the module outline, so the 20 butted joints are soldered and inspected from outside (F3-001; pcb_rules_check land table PASS). *(continued below)*
	REQ-025 evidence, continued (1/1): The ORIG label is etched left of JP1 beside the pad 2–1 gap; the hardware README describes that position (ADR-028 D-2), the manual still says 'below' (REQ-036). Module height and pin assumptions: first-build list in REQ-029. T: NOT RUN.			
REQ-026	**I2C rise time (PCB)** The PCB-build I2C bus shall meet the rise time for its clock: ≤ 300 ns at 400 kHz (≤ 1000 ns at 100 kHz), with the module at 0x5A.	A + T [PHYSICAL]	PARTIAL	A: PASS. R13/R14 4.7 kΩ: t_r 156 ns at 60 pF, C_b limit 115.2 pF (systems, with RPU 80 kΩ from S-17); electronics gives 103 ns at 38 pF. The 220 Ω series resistors add 2.2 ns (ASSUMPTION of 10 pF pad capacitance). IO levels with 220 Ω: B5–B8 PASS, minimum margin 0.138 V (S-17 Table 625, S-04 Table 5). T: NOT RUN (plan item P-4).

ID	Requirement	Verification	Status	Evidence
REQ-027	**Single-layer SMD PCB** The PCB shall have copper on one layer only, F.Cu (the component side), and carry surface-mount parts only. The only holes shall be the 4 M3 mounting holes. There shall be no vias and no copper pour. The only wire links shall be the 2 I2C configuration links (J1 SCL/SDA to the S-row pads of J3); JP1 is a solder bridge on etched pads.	I	PASS	I: PASS on the design file (T-104, T-103 board on fd7140b). pcb_rules_check `SUMMARY: 26 checks, 26 PASS, 0 FAIL`: 0 B.Cu objects, 0 vias, 0 copper zones (3 keep-out rule areas, the third for the Pico 2 TP7), 127 SMD pads on F.Cu with no drill, 121 F.Cu segments, 4 M3 NPTH holes on H1..H4 matching the 4 hits of the drill file, every multi-pad net one F.Cu island (SW1 pads joined inside the switch only), J1 only SCL/SDA. Every check was mutation-tested on a scratch copy (management/handoffs/systems-engineer/T-104/mutation_results.txt, `MUTATION SUMMARY: 85 proved, 0 not proved`). A design-file check: no board has been etched or drilled.
REQ-028	**Board outline** The board outline shall be 100 × 70 mm.	I	PASS	Outline 100.000 × 70.000 mm (sim/systems/pcb_rules_check.out, T-093: `PASS REQ-028 board outline 100 x 70 mm`). Board label 'FrutaMIDI v1.3' etched on F.Cu (ADR-023): interface_check `PASS pcb.label_v1.3`. The label is not part of this requirement's statement.

ID	Requirement	Verification	Status	Evidence
REQ-029	**Home-etch and SMD land rules** Home-etch rules: signal tracks ≥ 1.0 mm (≥ 0.8 mm for short stubs up to 6 mm), power tracks ≥ 1.2 mm, clearance between nets ≥ 0.6 mm (DRC hard minimum 0.5 mm), copper ≥ 0.5 mm from the board edge. SMD lands shall be ≥ 1.0 mm in both directions and match the v1.3 land table after T-103 (A1 1.6 × 4.2 mm, A2 1.8 × 4.2 mm electrode row and 1.8 × 5.515 mm S row, J2 3.0 × 4.5 mm, JP1 1.94 × 2.4 mm at pitch 3.81 mm, gap ≥ 1.8 mm, R1–R12 1.9 × 3.35 mm, R13–R16 1.775 × 1.8 mm, C1 1.875 × 1.8 mm, SW1 1.55 × 1.3 mm, RV1 3.0 mm, J1 2.2 mm, J3 1.8 mm), with no bare copper in the 3 Pico keep-out areas. The Pico end shall be inside the board edge and the USB receptacle face ≥ 1.0 mm beyond it (ADR-028 D-1). The remaining holes shall be drilled ≥ 0.8 mm with copper ≥ 0.5 mm from the hole edge. The annular-ring rule no longer applies because no plated hole remains.	I + A (DRC)	PASS	I + A PASS on the design file (T-104, T-103 board on fd7140b). pcb_rules_check `SUMMARY: 26 checks, 26 PASS, 0 FAIL`: track widths 0.8/1.0/1.2 mm, 0.8 mm only on 5 stubs (longest 5.08 mm), power ≥ 1.2 mm; minimum copper clearance between nets 0.680 mm (C1.2 to the SCL_ PICO track, MEASUREMENT, pads and tracks; etched text left to DRC); nearest copper 0.680 mm from the board edge at A1 pin 1 (rule 0.5 mm); land table after T-103 and ≥ 1.0 mm rule PASS (smallest 1.30 mm); 0 copper items in the 3 keep-out areas; Pico end 0.11 mm inside the edge and USB face 1.19 mm beyond it (`ADR-028 D-1`, limit 1.0 mm ASSUMPTION; accepted 0.11 mm short of flush, ADR-028 D-1); JP1 gap 1.87 mm; J2 pads 3.0 × 4.5 mm; hole clearance; spark gap (ADR-020) 1.400 mm on E0–E11. *(continued below)*

REQ-029 evidence, continued (1/1): Electronics check_smd (T-103) adds: every hand-soldered land ≥ 1.0 mm of copper outside the body or strip (A1 1.79, A2 1.73, SW1 1.65, R1 1.17, R13 1.05, C1 1.05 mm) and copper ≥ 0.6 mm from all 11 Pico 2 underside features (worst 2.14 mm) (MEASUREMENT on the design file, reports/electronics/evidence/T-103/). Every systems check was mutation-tested (management/handoffs/systems-engineer/T-104/mutation_results.txt). The rules do not prove that the lands can be etched or hand-soldered (R-003, R-013, R-014): user trial etch and solder are NOT RUN. First-build measurement list (T-103 build ASSUMPTIONS, each to be measured on the first build, [PHYSICAL], NOT RUN): (1) height of the supplied header strip, assumed 2.5 mm; (2) exposed pin length after cutting, set to 3.0 mm, all 20 equal; (3) Adafruit module board thickness, assumed 1.6 mm, giving the module underside about 5.5 mm and its top about 7.1 mm above the copper; (4) C1 body, assumed up to 3.4 × 1.8 mm (Samsung datasheet not opened), and the 1.0 mm copper beyond it; (5) B3S body 6.0 + 0.1 mm; (6) cut and transfer tolerance, assumed 1.0 mm (0.5 transfer + 0.5 trim): measure the USB receptacle face beyond the trimmed edge (design 1.19 mm, limit 1.0 mm) and that the Pico end is inside the edge (design 0.11 mm, ADR-028 D-1); (7) one RC2512 body L and termination I1 (REQ-023 step 1). If a value falls outside its assumption, re-run check_smd and pcb_rules_check with the measured value.

ID	Requirement	Verification	Status	Evidence
REQ-030	**Electrode wire pads** The PCB shall provide 14 solder-wire pads at 5.08 mm pitch: GND, E0..E11, GND. Pad k (k = 2..13) shall reach MPR121 electrode k − 2 through its series resistor.	I	PASS	interface_check `pcb.J2.T1..T14`, `pcb.A2.pin2..13`, `pcb.J2_solder_pads`, and pcb_rules_check `ADR-023 J2 is 14 SMD solder-wire pads` PASS (T-093). The project definition was amended for CR-003 (management/PROJECT_DEFINITION.md amendment of 2026-11-09, commit 410933d); its older screw-terminal line is superseded by that amendment. Strength of the wire joints and hot-melt relief is NOT RUN (R-013).
REQ-031	**Pin map consistency** The firmware PCB pin map (I2C1 SDA GP26 / SCL GP27, button GP22, pot GP28/ADC2) shall match the PCB nets. The breadboard map (I2C0 GP4/GP5, button GP15, pot GP26) shall match README.md.	I	PASS	interface_check: 0 pin mismatches.
REQ-032	**Core BOM cost** The core BOM (assembled board, excluding tools, consumables and USB cable) shall cost ≤ 30 EUR.	A	PASS	A: PASS on the basis ruled by ADR-027 D-1 (core BOM excluding shipping, consumables and tools, as in every earlier revision). Core 14.84–16.84 EUR ex VAT (CALCULATION, sim/systems/budgets.out revision 9, `PASS C1 core BOM upper bound <= 30 EUR (ex VAT, ex shipping) :: core 14.84 .. 16.84 EUR`), margin 13.16 EUR at the upper bound. With VAT at 23 % (ASSUMPTION rate) the core is 18.25–20.71 EUR, also ≤ 30 EUR (CALCULATION). Shipping: NOT CONFIRMED (no Mouser checkout opened); outside the ruled basis and reported here, not included. Includes: the Mouser lines 13.84 EUR for one board (A1 SC0915 3.44, A2 1982 6.84, R1–R12 RC2512FK-073K3L 12 × 0.109 = 1.31 at the 10+ tier (1 pc 0.189), R13–R16 RC1206 4 × 0.086, C1 CL31B104KBCNNNC 0.12, SW1 B3S-1000 0.697, RV1 PTV09A-4015F-B103 0.765, knob SPK-016A 0.327; *(continued below)*

ID	Requirement	Verification	Status	Evidence
	REQ-032 evidence, continued (1/1): SOURCE Mouser Portugal 2026-11-11, EUR ex VAT, T96-S0b/T96-S2/T96-S5 via S-48) and the copper clad 1.0–3.0 EUR (ASSUMPTION). Excludes: shipping, consumables (etchant, toner, solder), wire, hot-melt, epoxy, USB cable, optional crocodile leads and M3 hardware, tools (ADR-007 D-7, ADR-027 D-1). Plan B (T-097, R-018) would add 6.18 EUR: 21.02–23.02 EUR, still $\leq$ 30 EUR (CALCULATION). Availability: Mouser PT stock 13 for 12 needed (one spare), accepted by ADR-027 D-3 (risk R-016). The earlier FAIL (20.10–53.55 EUR, revision 8) came from the unpriced HPWC 3k0 part, replaced by ADR-026.			
REQ-033	**Firmware build matrix** The firmware shall build without errors for Pico, Pico W, Pico 2 and Pico 2 W, each for the breadboard and PCB variants (FQBN `rp2040:rp2040:<board>:usbstack=tinyusb`, plus `-DFRUTAMIDI_PCB` for PCB), and for `arduino:avr:leonardo` (optional).	T (compile)	PASS	`reports/firmware/T-044/build/*.log`: 9 of 9 logs (Pico, Pico W, Pico 2, Pico 2 W × std/PCB, plus Leonardo) contain 'Sketch uses'. A clean compile is not a hardware validation.
REQ-034	**Browser sampler** The browser sampler `docs/index.html` shall work in Chrome and Edge (Web MIDI) on Windows and macOS. It shall map its 12 pads to notes 36–47 by default, treat Note On with velocity 0 as Note Off, send unmapped notes to the synthesiser, apply aftertouch (0xA0) to the filter, and offer a keyboard fallback and dropped-file samples.	T	PARTIAL	Automated browser tests with a stubbed Web MIDI PASS (T-020, T-021 handoffs). A real browser, a real device and audible output are NOT RUN.
REQ-035	**English language** All project files, code, comments, reports, generated PCB text, fabrication file names and user-visible strings shall be in English.	I	PASS	I: 0 Portuguese-word matches in firmware (`.ino`, `fm_*.h`), gen_*.py, docs/index.html, README.md, hardware/README.md and docs/manual/MANUAL.md. Etched copper text and assembly-map text are English (SCALE, OCTAVE, ORIG, QT, legend; hardware/frutamidi.kicad_pcb gr_text, T-093); fabrication file names are English; no broken relative links (`doc.links.*` PASS).

ID	Requirement	Verification	Status	Evidence
REQ-036	**User and assembly manual** The user and assembly manual shall match the current PCB. It shall cover both module revisions, the JP1 selector and data wires, flashing, first start, a valid pre-power continuity checklist and troubleshooting.	I	PASS	Inspection by interface_check on the T-103 board with the T-105 documentation (T-106, HEAD f6693c9 plus the uncommitted T-103/T-105 working tree): `SUMMARY documentation: 31 checks, 31 PASS, 0 FAIL`, `SUMMARY: 121 checks, 121 PASS, 0 FAIL`. docs/manual/MANUAL.md: `doc.manual.matches_v1.3` and `doc.manual.JP1_label_position` PASS (ORIG etched left of JP1 beside the pad 2–1 gap, ADR-028 D-2); the manual now covers the supported boards (Pico SC0915 and Pico 2 only), board trimming at the USB end, plug-seat and rocking checks, module mounting (pins 3.0 mm, solder from outside) and the post-solder checks (pads 36/39/40 not to GND, BTN not to pad 30) (T-105 handoff; read by systems at lines 39, 43, 186–190). *(continued below)*
	REQ-036 evidence, continued (1/1): README.md (root): `doc.root_readme.matches_v1.3` and `doc.root_readme.pcb_variants` PASS (Pico W / Pico 2 W PCB rows marked not supported on the v1.3 PCB). hardware/README.md: all `doc.hw_readme.*` checks PASS. Every documentation check was mutation-tested (management/handoffs/systems-engineer/T-106/mutation_results.txt, `MUTATION SUMMARY: 86 proved, 0 not proved`). The PASS covers consistency between the documents and the design file only: readability and the hand-soldering steps are not tried, and nothing has been built.			
REQ-037	**Sensing range (physical)** [PHYSICAL] Touch detection shall work on objects from a bare pad (about 20 pF) to large fruit or a glass of water (up to about 500 pF, ASSUMPTION), with leads of 20–50 cm, after recalibration.	T [PHYSICAL]	NOT RUN	A: the MPR121 can charge up to about 1203 pF (CALCULATION). Latency targets hold up to 210 pF at 400 kHz. AUTOCONFIG limits USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (0x7C D7) are now SOURCE (ADR-010/011, MPR121 datasheet S-006 in management/SOURCES.md). T: NOT RUN.

ID	Requirement	Verification	Status	Evidence
REQ-038	**False triggers and crosstalk (physical)** [PHYSICAL] There shall be no false Note On in 10 min idle with all 12 electrodes connected to objects, and no crosstalk when touching a neighbouring electrode.	T [PHYSICAL]	NOT RUN	T: NOT RUN [PHYSICAL]. Plan addition (C2-012, updated T-093): after the T-094 re-route the nearest I^2C copper is 3.30 mm from electrode copper (SDA track#99 at y 70.90 mm to A2.10 /ELE8, MEASUREMENT on the design file, pcb_rules_check INFO, T-099; it was 0.65 mm in T-093). The EE estimate after T-103 is 36.6 fF SDA-to-pad and a 6.72 mV = 2.09 counts step against TOUCH_THRESHOLD 12 (CALCULATION, calc_i2c_coupling.out; 1.89 counts in T-094), inside the 3-count criterion (budgets D2 PASS; risk R-017 Low). Test: with all 12 electrodes connected and untouched, log the MPR121 filtered data and baseline of each electrode (a) with normal 400 kHz polling and (b) with I^2C traffic reduced to one read per 100 ms (debug build); proposed criterion: the peak-to-peak difference between (a) and (b) is below half RELEASE_THRESHOLD, i.e. 3 counts (ASSUMPTION; *(continued below)*
	REQ-038 evidence, continued (1/1): RELEASE_THRESHOLD 6 and TOUCH_THRESHOLD 12 are SOURCE firmware/FrutaMIDI/FrutaMIDI.ino), and no false Note On in 10 min. Shares setup with plan item P-4.			
REQ-039	**JP1 cannot short 3V3** No position of the JP1 3V3 selector (ORIG, QT, open) shall connect +3V3 to GND.	I	PASS	S1 and S7 are never GND in either revision (`pcb.JP1_no_gnd_reachable`).

ID	Requirement	Verification	Status	Evidence
REQ-040	**JP1 wrong-position tolerance** A wrong JP1 position (ORIG on a STEMMA QT module, or QT on an original module), or both sides bridged, shall not apply a voltage outside the MPR121 absolute maximum ratings (inputs $\leq$ VDD + 0.3 V, S-04 Table 3), and shall not short +3V3 through a module output.	A	FAIL	A: still FAIL as stated. The ORIG bridge on a QT module puts +3V3 on ADDR, and the QT bridge on an original module puts it on IRQ, both above VDD + 0.3 V on an unpowered module. Both sides bridged on an original module ties +3V3 to the IRQ open drain. Mitigation per ADR-009 D-10: etched ORIG/QT labels and assembly-map legend (`pcb.JP1_legend` PASS; the longer legend is no longer on the board, ADR-023), manual warning, and the firmware 'sensor not found' LED pattern. Residual risk R-007 accepted by the manager; ADR-027 D-5 accepts the v1.3 form of the legend (etched ORIG/QT on F.Cu plus the assembly map on F.Fab, no silkscreen). The status stays FAIL against the statement: the hazard is accepted, not removed.
REQ-041	**I2C rise time (breadboard, with pull-ups)** The breadboard-build I2C bus, with the two 4.7 kΩ pull-ups required by ADR-009 D-11, shall meet the rise time for 400 kHz.	A + T [PHYSICAL]	PARTIAL	A: PASS. t_r 156 ns at 60 pF with 4.7 kΩ (CALCULATION). Without them, 452 ns: FAIL above 39.8 pF. I: README.md lists the two 4.7 kΩ pull-ups in its BOM and wiring table. T: NOT RUN (plan item P-4).
REQ-042	**Shared-note reference counting** When two or more pads map to the same MIDI note, the note shall be sent once (Note On on the first pad), stay sounding while any of those pads is held, and get its Note Off only when the last of them is released (reference counting). Aftertouch for that note shall be the maximum of the holding pads' values, with a final pressure of 0 on the last release.	T (host) + T [PHYSICAL]	PARTIAL	Host tests PASS: `tests/firmware/logic/test_notes_health.py` (reviewer sequence, `releaseAll`), `firmware/FrutaMIDI/fm_notes.h` (T-044, 295 passed). Physical NOT RUN.

Status counts (42 requirements): PASS 18, PARTIAL 18, FAIL 2, NOT RUN 4.

3 PCB v1.3: single-sided SMD board (CR-003)

Inputs: branch `change/CR-003` at `f6693c9`. Sources:

- change request and decisions: [`changes/CR-003.md`], [`management/DECISIONS.md`] (ADR-023 to ADR-028);
- electronics analysis section 8: [`reports/electronics/analysis.md`];
- handoffs: [`management/handoffs/electronics-engineer-T-103.md`], [`management/handoffs/systems-engineer-T-104.md`];
- review: [`reports/review/CR-003/findings.md`];

- budgets and metrics: [`sim/systems/budgets.out`], [`management/metrics.json`].

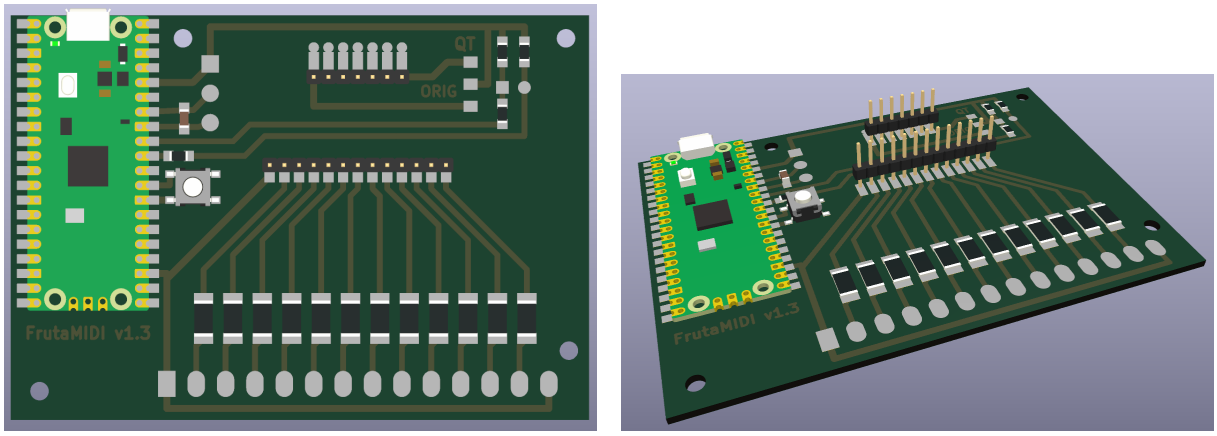
This section describes the current board and supersedes the v1.2 board-specific content of Sections 4 to 11 where they differ.

Status. Every result in this section is a design-file check, a calculation or a simulation. **The v1.3 board has not been etched, soldered, powered or measured, and nothing in it is hardware-validated.** ERC, DRC and the rules checks passing do not validate the hardware.

3.1 Reasons for the change

The user makes boards at home by chemical etching and does not want to drill them (CR-003, 2026-11-09). V1.2 had 133 drilled holes (MEASUREMENT, drill file, [`changes/CR-003.md`]); v1.3 has 4, the optional M3 mounting holes (MEASUREMENT, `v1.3_drill_holes_total` in [`management/metrics.json`]). The conversion (ADR-023, ADR-026):

- one copper layer, on the component side (F.Cu); surface-mount parts only; no vias, no copper pour, no silkscreen (the ORIG/QT legend is etched in copper and the rest is on the assembly map);
- the Pico (A1) is soldered on its castellations over an etched bare-copper keep-out; the Adafruit 1982 module (A2) stands on its own header pins, cut short and soldered onto surface lands; no sockets, so a failed Pico or module has to be desoldered from the surface (accepted, ADR-023);
- R1–R12 become 2512 chip resistors, R13–R16 and C1 become 1206, SW1 is the SMD B3S-1000, RV1 moves off the board on 3 wire pads (ADR-023 D-2), and J2 becomes 14 solder-wire pads with hot-melt strain relief (ADR-023 D-3);
- JP1 is a 3-pad surface solder bridge and J1 two surface pads for the I^2C configuration wires; a bare MPR121 was rejected (0.40 mm pitch against the 0.6 mm clearance rule, SOURCE, ADR-023);
- the firmware and the pin map do not change (REQ-031 PASS).



(a) top view (b) 3D view

Figure 1: PCB v1.3 after T-103, KiCad renders, not photos (`hardware/fabrication/images/board-top.png` and `board-3d.png`). The Pico USB end is at the top edge; R1–R12 and the J2 wire pads are along the bottom. The module itself is not modelled: only its header pins are drawn, raised 3.0 mm (the cut-pin length, ASSUMPTION), standing on the two rows of lands in the centre.

3.2 Board geometry after the review fixes (T-103)

Table 2: v1.3 board values after T-103 (from `management/metrics.json`).

Quantity	Value	Label
Board outline	100 × 70 mm	MEASUREMENT (REQ-028)
Pico (A1) anchor	y = 1.48 mm	MEASUREMENT (design file)
Pico end inside the board edge	0.11 mm	CALCULATION
USB receptacle face beyond the edge	1.19 mm (limit 1.0 mm)	CALCULATION from SOURCE 1.30 mm; limit ASSUMPTION
Cutting tolerance	1.0 mm (0.5 transfer + 0.5 trim)	ASSUMPTION
A2 lands, electrode row / S row	1.8 × 4.2 mm / 1.8 × 5.515 mm	MEASUREMENT (design file)
A2 lands beyond the module outline	1.73–1.775 mm	MEASUREMENT (design file)
Header strip, cut pin, module board	2.5 / 3.0 / 1.6 mm	ASSUMPTION (first-build list)
Module underside / board top above the copper	5.5 / 7.1 mm	CALCULATION on the ASSUMPTIONs above
1206 lands, R13–R16 / C1	1.775 × 1.8 / 1.875 × 1.8 mm	MEASUREMENT (design file)
Keep-out rule areas under the Pico	3	MEASUREMENT (design file)
Copper to the nearest Pico 2 underside feature	2.14 mm (11 features)	MEASUREMENT (design file)
Minimum copper clearance between nets	0.680 mm (rule 0.6 mm)	MEASUREMENT (design file)
Minimum copper to board edge	0.680 mm (rule 0.5 mm)	MEASUREMENT (design file)
JP1 pad gap	1.87 mm	MEASUREMENT (design file)
Electrode pad to GND edge track (spark gap)	1.401 mm, all 12	MEASUREMENT (design file)
SDA track to nearest electrode pad	3.30 mm	MEASUREMENT (design file)

The Pico sits 0.11 mm short of flush with the edge: flush would put the pin 1 and pin 40 (VBUS) lands 0.495 mm from the etched outline line, below the 0.6 mm rule (CALCULATION); the manager accepted this (ADR-028 D-1). The PCB supports only the Raspberry Pi Pico (SC0915, recommended) and the Pico 2; Pico W and Pico 2 W are not supported until their undersides are checked (ADR-028 D-3).

Checks on the design files (T-103, T-104, T-105).

- ERC: "ERC messages: 0 Errors 0 Warnings 0"; DRC: "Found 0 DRC violations", "Found 0 unconnected pads"; "Found 0 schematic parity issues" (KiCad 10.0.4, T-103 handoff).
- `check_smd`: "SUMMARY: 65 checks, 65 PASS, 0 FAIL" (T-103); the new checks were each shown to fail on a mutated copy.
- `pcb_rules_check`: "SUMMARY: 26 checks, 26 PASS, 0 FAIL"; mutation tests "MUTATION SUMMARY: 85 proved, 0 not proved" (T-104).

- `interface_check` after the T-105 documentation fixes: "SUMMARY: 121 checks, 121 PASS, 0 FAIL" [`management/handoffs/documentation-engineer/T-105/interface_check.out`].
- `budgets`: "SUMMARY: 35 checks, 31 PASS, 4 FAIL"; the 4 FAILs are A3b (VBUS, REQ-018) and the three 100 kHz latency rows, unchanged from earlier revisions (T-104).

3.3 Bill of materials and cost

All parts are bought from Mouser (ADR-024). Prices are SOURCE Mouser Portugal, 2026-11-11, EUR excluding VAT [`sim/systems/budgets.out`].

Table 3: v1.3 core BOM (budgets revision 10).

Ref.	Part (Mouser number)	Qty	EUR, label
A1	Raspberry Pi Pico SC0915 (358-SC0915)	1	3.44 SOURCE
A2	Adafruit MPR121 breakout 1982 (485-1982)	1	6.84 SOURCE
R1–R12	YAGEO RC2512FK-073K3L, 3k3 1 %, 2512 (603-RC2512FK-073K3L)	12	1.31 at 0.109 (10+ tier) SOURCE
R13–R16	YAGEO RC1206FR-074K7L and RC1206FR-07220RL	4	0.34 SOURCE
C1	Samsung CL31B104KBCNNNC 100 nF 1206 (187-CL31B104KBCNNNC)	1	0.12 SOURCE
SW1	B3S-1000 (653-B3S-1000; ARATAS, formerly Omron Components)	1	0.70 SOURCE
RV1	Bourns PTV09A-4015F-B103 (652-PTV09A-4015FB103), off-board	1	0.77 SOURCE
Knob	Same Sky SPK-016A (179-SPK-016A); bore fit NOT CONFIRMED	1	0.33 SOURCE
—	Single-sided copper clad 100 × 70 mm	1	1.00–3.00 ASSUMPTION
	Mouser lines only		13.84 CALCULATION
	Core BOM, ex VAT, ex shipping		**14.84–16.84** CALCULATION
	Core BOM with 23 % VAT (rate ASSUMPTION)		18.25–20.71 CALCULATION
	Shipping		NOT CONFIRMED

J1, J2, J3 and JP1 are etched copper only and are excluded from the BOM; the knob has no schematic symbol and is listed in the assembly guide only (ADR-028 D-4; budgets C2 PASS, "differences []"). REQ-032 ($\leq$ 30 EUR) is PASS on the basis ruled by ADR-027 D-1 (core BOM excluding shipping, consumables and tools), with shipping reported as NOT CONFIRMED.

Supply. Mouser showed 13 RC2512FK-073K3L in stock for the 12 needed, next delivery 2027-05-18 (SOURCE, 2026-11-11; risk R-016, accepted by ADR-027 D-3). The named alternative is Vishay CRCW25123K30FKEG: 12 × 0.424 = 5.09 EUR, +3.78 EUR, core 18.62–20.62 EUR (CALCULATION from SOURCE 0.424 EUR). Its datasheet has not been opened, so its body size is NOT CONFIRMED; it may be used only after `check_gap` (d) and `sim_electrode` are re-run with its values (ADR-028 D-5).

How the resistor was chosen. ADR-023 chose TT Electronics HPWC2512 3k0, which could not be bought in small quantities; ADR-024 chose the Vishay CMB 0207 (one-board quote about 17.6 EUR, CALCULATION), with an accepted 3.2 mm conductor gap; ADR-025 chose the HPWC2512-3K3JT1 (one-board quote about 27.5 EUR, CALCULATION); ADR-026, the user's final decision, chose the standard YAGEO part and accepted that its pulse capability is not proven.

3.4 ESD analysis of the electrode path

The protection target is an IEC 61000-4-2 contact discharge of ±4 kV, itself an ASSUMPTION (level 2; the standard was not read). The model is a 150 pF / 330 Ω source into the 3.3 kΩ series resistor [`reports/electronics/analysis.md`] section 8 [`sim/electronics/calc/calc_pulse_rated.out`].

Table 4: ESD numbers for R1–R12 = RC2512FK-073K3L (analysis.md section 8).

Quantity	Value	Label
Ratings: max working / overload voltage	200 / 500 V; no pulse, surge or ESD rating published	SOURCE YAGEO RC_L V.11
Voltage across R at 4 kV / 8 kV	3636 / 7273 V (7.3× the overload rating at 4 kV)	CALCULATION
Energy into R per strike, 4 kV / 8 kV	1.09 / 4.36 mJ, τ = 0.544 µs	CALCULATION
Peak power, 4 kV / 8 kV	4007 / 16028 W	CALCULATION
Old 1.2/50 µs acceptance energy (MBE0414 182 mJ; HPWC 231 mJ)	167× / 212× one 4 kV strike	CALCULATION
Film temperature rise, 4 kV / 8 kV	3.1 / 12.6 K; 31 / 126 K with a ×10 trim hot spot	CALCULATION on ASSUMPTION
Sensor pin current at 4 kV	1.10 A (1.11 A at 3267 Ω) vs 1.33 A inferred limit; margin 1.20–1.22	CALCULATION; limit ASSUMPTION
Charge, IEC 4 kV vs HBM 2 kV	600 vs 200 nC (3.0×)	CALCULATION
Decay time, IEC through 3.3 kΩ vs HBM	0.54 vs 0.15 µs (3.6×)	CALCULATION
Pin current at 8 kV	2.20 A: FAIL	CALCULATION
Copper gap across R / termination gap	5.00 mm / 4.65 mm (4.08 mm at ±0.5 mm placement)	MEASUREMENT (design file) / CALCULATION
Gap rule at 4 kV / 8 kV	≥ 3.5 / ≥ 7.5 mm: PASS / FAIL	CALCULATION on ASSUMPTION models
Electrode sensitivity, worst ratio at 3267–3333 Ω	0.904 (limit 0.90)	SIMULATION

What the numbers show and do not show.

- **Pulse capability of the resistor: NOT PROVEN** at 4 kV and 8 kV. The maker publishes no pulse rating. The energy comparison (one 4 kV strike is 1/167 of the energy the earlier, rated part was accepted against) is not a rating for this part: a thick-film resistor under a fast pulse fails at the local field and hot spot of the laser trim, which the datasheet does not cover. The user accepted this to keep the board cheap (ADR-026).
- The pin-current margin of 1.20–1.22 compares peak current only. The 4 kV IEC pulse carries 3.0 times the charge and lasts 3.6 times longer than the HBM pulse the sensor rating comes from, so the margin is not a proof (review F3-005).
- The spark gap of 1.401 mm from each electrode pad to the GND edge track is intended to take a strong discharge before the 5 mm gap around the resistor (ADR-020); it is also unproven.
- Only an ESD-gun test can close REQ-023 (PARTIAL, test NOT RUN).

Plan B and its trigger (ADR-027 D-2, risk R-018). If, after the physical ESD series, any R1–R12 reads outside 3234–3366 Ω (±2 %, ASSUMPTION window from the 0.901 sensitivity at 3366 Ω, SIMULATION), a channel stops detecting touch, or an arc crosses a resistor, the board moves to the T-097 clamp: 12 × Bourns CDSOD323-T05LC to GND with R1–R12 at

1 kΩ (sensitivity 0.938, SIMULATION; +6.18 EUR, core 21.02–23.02 EUR, CALCULATION), after its maximum capacitance and leakage are confirmed.

Board-level residual items (ADR-028 D-7). Each rests on ASSUMPTION inputs and is NOT RUN [PHYSICAL]:

- GND bounce between the module and the Pico during a strike to the spark gap: up to 330 V for about 1 ns; 1.49 A and 0.90 nC per I^2C line, against 2.67 A and 400 nC for the RP2040 HBM 4 kV FT-pin equivalent (CALCULATION; budgets D1 PASS; the GP26/GP27 rating is NOT CONFIRMED, risk R-010). The taller module pin path of T-103 raised it from 236 V.
- SDA coupling to the electrode pads: 36.6 fF, a 2.09-count step against the 12-count touch threshold (CALCULATION, analysis.md section 8.5; budgets D2 PASS against the 3-count proposal of REQ-038; risk R-017).
- USB pull: 35.6 N on the most loaded M3 screw at a 30 N pull (CALCULATION; budgets D3 PASS against an unsourced 100 N ASSUMPTION; risk R-013).

3.5 Independent review T-102 and dispositions

The independent reviewer reviewed `9505245` and found the board **not ready for the first home-etched build**: F3-001 blocked, F3-002 and F3-003 had to be fixed or scoped first [`reports/review/CR-003/findings.md`]. The fixes are T-103 (electronics), T-104 (systems) and T-105 (documentation); the manager's rulings are ADR-028. Status below is the result of the check named for each fix on the design files or documents; a re-review of the fixes is NOT RUN (ADR-028 D-8).

Table 5: CR-003 review findings and dispositions (ADR-028).

ID	Sev.	Finding	Disposition	Status
F3-001	critical	A2 lands entirely under the header strip and module outline; joints cannot be reached or inspected.	Lands lengthened outward to 1.73–1.775 mm beyond the outline; strip pushed against the module, pins cut to 3.0 mm, soldered from outside (T-103; check_smd land check).	PASS (design file)
F3-002	major	USB face only 0.67 mm beyond the edge.	Pico moved; face 1.19 mm beyond, Pico end 0.11 mm short of flush, accepted (ADR-028 D-1); oversize cut, trim and plug-seat check in the guide.	PARTIAL
F3-003	major	Pico 2 TP7 outside the keep-outs; Pico W / Pico 2 W never checked.	TP7 keep-out added, 11 Pico 2 features clear by $\geq$ 2.14 mm; PCB scoped to Pico and Pico 2 (ADR-028 D-3); README rows corrected in T-105.	PASS (design file, documents)
F3-004	major	ESD analysis described the old HPWC part.	analysis.md section 8 on RC2512 (pulse NOT PROVEN); 7.1 and 7.2 marked superseded; guide link fixed.	PASS (document)
F3-005	minor	Pin margin compares peak current only.	Charge 3.0× and decay 3.6× stated beside the margin.	PASS (document)
F3-006	minor	ORIG label is left of JP1, not below.	Copper unchanged; guide (T-103) and manual (T-105) describe the real position (ADR-028 D-2).	PASS (documents)
F3-007	minor	No post-solder short checks for VBUS/VSYS and BTN/RUN.	Checks added: pads 36, 39, 40 not to GND; BTN not to pin 30.	PASS (document)
F3-008	minor	Pin-1 mark level with pin 2.	Mark moved level with pin 1, square drawn in pad 1.	PASS (render inspected)

ID	Sev.	Finding	Disposition	Status
F3-009	minor	Records disagree (REQ-032, check counts, stock date, INTERFACES).	Requirements draft-13/14 and INTERFACES frozen-9 proposal (T-104); ADR-026 stock date superseded (ADR-028 D-6); frozen-9 not yet frozen.	PARTIAL
F3-010	minor	J1 in the BOM without MPN; knob outside the KiCad BOM; SW1 maker.	J1, J2, J3, JP1 copper only; knob in the guide only; SW1 genuine B3S-1000 (ADR-028 D-4).	PASS (budgets C2)
F3-011	info	Wire direction and strip length at J2.	Wires leave upward, strip 3–4 mm.	PASS (document)
F3-012	info	Solder at the USB shell pins can tilt the Pico.	Rocking check before tacking.	PASS (document)
F3-013	info	"module" used for the Pico.	Wording fixed.	PASS (document)
F3-014	info	1206 lands 0.43 mm beyond the body.	Lands lengthened; $\geq$ 1.05 mm beyond the body.	PASS (design file)
F3-015	info	Regeneration not byte-identical (UUIDs).	No action; geometry identical.	—

3.6 First-build measurement list

The first build has to measure the values the design rests on. All items are NOT RUN [PHYSICAL]; sources are the REQ-029 and REQ-023 evidence in [`requirements/requirements.csv`] and the assembly guide [`hardware/README.md`]. If a value falls outside its assumption, `check_smd` and `pcb_rules_check` are re-run with the measured value.

Table 6: v1.3 first-build measurements (all NOT RUN).

#	What	Design value (label)	How / criterion
FB-1	Height of the supplied header strip	2.5 mm (ASSUMPTION)	Caliper; re-run the checks if different
FB-2	Exposed pin length after cutting, all 20	3.0 mm (ASSUMPTION)	Caliper; all equal, module level on both rows
FB-3	Module board thickness; module underside and top above the copper	1.6 mm (ASSUMPTION); 5.5 / 7.1 mm (CALCULATION)	Caliper
FB-4	C1 body, and copper beyond it	up to 3.4 × 1.8 mm (ASSUMPTION); $\geq$ 1.0 mm	Caliper on the fitted part
FB-5	B3S body	6.0 + 0.1 mm (ASSUMPTION)	Caliper
FB-6	USB receptacle face beyond the trimmed edge; Pico end inside the edge	1.19 mm (limit 1.0 mm); 0.11 mm (CALCULATION)	Caliper after trimming; plug fully seated before soldering
FB-7	One RC2512 body L and termination I1	L 6.35 ±0.10, I1 0.60 ±0.20 mm (SOURCE)	Caliper; inside 6.25–6.45 and 0.40–0.80 mm, else re-run `check_gap` (d)
FB-8	R1–R12 before soldering and after the ESD series	3267–3333 Ω (3.3 kΩ ±1 %, CALCULATION)	Ohmmeter; after the series every part within 3234–3366 Ω, else plan B
FB-9	Pico flat before tacking	—	Must not rock at any corner
FB-10	Post-solder continuity	—	Pico pads 36, 39, 40 not shorted to GND; SW1 BTN not to pad 30; JP1 centre only to S1 (ORIG) or S7 (QT)

#	What	Design value (label)	How / criterion
FB-11	ESD series ±4 kV contact on E0–E11 and the RV1 knob	—	REQ-023 plan steps (1)–(3); arc path, resets and resistor values recorded
FB-12	SDA coupling step, GND bounce effects, USB pull	2.09 counts; 330 V / 1 ns; 35.6 N (Section 3.4)	Touch data during I^2C traffic; behaviour during strikes; no pad lift at a 30 N pull

3.7 Requirement status for CR-003 (draft-14)

Requirements draft-14 (T-106), 42 requirements: PASS 18, PARTIAL 18, FAIL 2 (REQ-018, REQ-040), NOT RUN 4 [`requirements/requirements.csv`]. Rows touched by CR-003:

Table 7: CR-003 requirements in draft-14.

ID	Title	Status	Basis
REQ-023	Electrode ESD robustness	PARTIAL	Analysis only; resistor pulse capability NOT PROVEN (ADR-026); ESD test NOT RUN
REQ-025	Both module revisions	PARTIAL	JP1 and wires checked on the design file; test NOT RUN
REQ-027	Single-layer SMD PCB	PASS	Design file (pcb_rules_check 26/26); no board etched
REQ-028	Board outline	PASS	100.000 × 70.000 mm on the design file
REQ-029	Home-etch and SMD land rules	PASS	Design file only; etching and hand soldering NOT RUN; first-build list above
REQ-030	Electrode wire pads	PASS	Design file
REQ-031	Pin map consistency	PASS	interface_check, 0 pin mismatches
REQ-032	Core BOM cost	PASS	14.84–16.84 EUR ex VAT, ex shipping (ADR-027 D-1); shipping NOT CONFIRMED
REQ-036	User and assembly manual	PASS	Draft-14 (T-106): interface_check 121/121, documentation 31/31, after T-105. Covers consistency of the manual and READMEs with the design files only; the hand-assembly steps have not been tried
REQ-040	JP1 wrong-position tolerance	FAIL	Accepted risk R-007 (ADR-027 D-5)

A PASS in this table means the analysis, inspection or design-file check passed. None of these requirements has been verified on a built board.

4 Architecture and decisions

v1.3 note. This section was written for v1.2. In v1.3 the electrodes connect through solder-wire pads (J2) instead of screw terminals, and the potentiometer is off the board; the signal chain, the sensor, the bus and the firmware are unchanged (Section 3).

4.1 System overview

Touch path: electrode object → lead → screw terminal → 3.3 kΩ series resistor (PCB) → MPR121 electrode pin; the MPR121 reports touch bits and filtered/baseline data over I^2C at 400 kHz to the Pico, which sends USB MIDI 1.0 messages to the host (browser sampler or DAW) [`management/INTERFACES.json`] [`reports/systems/architecture_options.md`].

4.2 Architecture options

Three options were compared on the same criteria [`reports/systems/architecture_options.md`]; option A was chosen (ADR-006).

Criterion	A: Pico + MPR121 (chosen)	B: Pico native sensing	C: Pro Micro + MPR121
Core BOM cost	23.60–34.60 EUR (CALCULATION, budgets.out, revision 1)	14.20–22.20 EUR (CALCULATION from ASSUMPTION prices)	22.60–34.60 EUR (ASSUMPTION clone price)
Sensing with fruit/water	Good; hardware autoconfiguration and baseline filter	Unproven; filtering in firmware	Same as A
Latency floor	4 ms at the sensor	Potentially below 1 ms (ASSUMPTION)	Same as A
Constraint compliance	Uses the fixed bought-in parts	Drops the Adafruit 1982 (definition change)	Drops the Pico family (definition change)
Risk level	Low–medium	High	Medium

The BOM figure for A above is the T-008 value; the v1.2 core BOM was 23.98–38.62 EUR (CALCULATION, budgets revision 7 after CR-002 and ADR-021); the v1.3 core BOM is 14.84–16.84 EUR excluding VAT and shipping (CALCULATION, budgets revision 10, Section 3.3) [`reports/systems/budgets.md`].

4.3 Decisions

Summary of [`management/DECISIONS.md`] (ADR-000 to ADR-028). Decisions taken by the manager under ADR-001 are flagged there for later user review.

ADR	Decision	Basis / evidence
ADR-000	Project folder, all files in English, existing paths kept, commit rules.	User, Phase A
ADR-001	Run gates G1–G4 end to end; publishing a release still needs explicit user approval.	User
ADR-002	Firmware is the source of truth for the long press: docs state 0.8 s (firmware `> 800` ms).	T-030 handoff
ADR-003	No "safe to touch" claim without evidence; wording follows the electronics analysis.	T-050
ADR-004	English file and folder names (`hardware/fabrication/`, `images/`).	T-051
ADR-005	Sampler: learning a used note swaps the two pads; notes flash before audio is enabled; omni channel; English UI and silkscreen.	T-020
ADR-006	Architecture option A (Pico + Adafruit MPR121 breakout); MPR121 sets a ≥ 4 ms sensing floor.	architecture_options.md
ADR-007	Latency ≤ 15 ms worst case and jitter ≤ 10 ms (relaxed, flagged); burst read M1; 24 pads breadboard only; JP1 3-pad selector replaces the 3V3 wire; hosts; BOM target ≤ 30 EUR; English.	T-008
ADR-008	3.3 kΩ series resistor per electrode (recorded as PASS ±4 kV, later downgraded to an estimate by review F-002; PARTIAL 8 kV, risk R-006); keep 400 kHz with 4.7 kΩ pull-ups; label v1.1 (superseded by ADR-019 D-2).	analysis.md

ADR	Decision	Basis / evidence
ADR-009	PARTIAL policy; 220 Ω I^2C series resistors (fault current ≤ 15 mA); JP1 silkscreen and warnings (R-007); breadboard 4.7 kΩ pull-ups.	T-010
ADR-010	Run-based burst reads (12 pads 945 µs vs 2430 µs, CALCULATION); AUTOCONFIG limits 200/180/130 at design VDD 3.2 V; SCTS = 0.	T-041
ADR-011	MPR121 register facts confirmed from the datasheet (S-006); three firmware ASSUMPTIONs become SOURCE.	T-042, manager
ADR-012	No datasheet-rated pulse resistor found; MFR-12 3k3 kept, REQ-023 PARTIAL, candidate CR-001 (superseded by CR-002 and ADR-019 D-1); R15/R16 220 Ω accepted.	T-054, T-055
ADR-013	Interfaces frozen as frozen-1; REQ-040 stays FAIL with accepted risk R-007.	Manager
ADR-014	All 16 review findings of 72f67f0 dispositioned (13 FIX, 1 ACCEPTED, 2 info FIX); fixes made on main before the baseline; no frozen interface changed.	Review T-060
ADR-015	BOM part numbers: J2 Phoenix Contact 1715721 (×7); RV1 named by series (RK09K, order code not sourced, PARTIAL accepted); female headers generic; SW1 Omron B3F-1000.	T-056
ADR-016	REQ-018 stays FAIL with a documented deviation (low-power USB port 4.40 V below the Pico's recommended 4.50 V; guide recommends a standard port or powered hub); documentation-only interface change applied as frozen-2.	T-012
ADR-017	Firmware review fixes: per-note reference counting (F-004), sensor health with loss detection and retry (F-005), `setClock` after every sensor `begin()` (F-006).	T-044
ADR-018	Re-review dispositions: F-007 fixed (build logs committed); N-001 accepted as a REQ-016 exception (lost-sensor restart 16.8–22.1 ms, fault path only); N-002 evidence refreshed; N-003 interfaces frozen-3 (IF-PCB-AUX).	T-061
ADR-019	CR-002 decisions. D-1: ESD scope stays ±4 kV; R1–R12 become Vishay MBE04140C3301FC100 (DIN0414, 3k3 1 %), 4.08 kV 1.2/50 µs rating at 3.3 kΩ (SOURCE EE-S17b), margin 1.12 at 4 kV (CALCULATION); 8 kV stays risk R-006, LSR37 deferred. D-2: board label "FrutaMIDI v1.2". D-3: cost increase accepted, REQ-032 stays FAIL. D-4: relayout accepted (module, J1, JP1, R13–R15, H4 moved; nets and pinout unchanged).	T-060, T-070
ADR-020	CR-002 re-review dispositions: end-cap gap risk (C2-002) controlled by forming the leads on a gauge and a caliper check in the physical plan; the E-pad to GND edge track stays as an intentional spark gap (C2-005); H3 moved to (96, 38) mm.	T-073
ADR-021	Manager PCB design review follow-up: JP1 pitch 3.81 mm (gap 1.87 mm); J2 pads 3.0 × 4.5 mm oval; RV1 tabs on AGND; C1 100 nF wiper filter; silkscreen overlaps removed; C2 1 µF rejected. Addendum: C1 cost accepted (REQ-032 stays FAIL); INTERFACES frozen-6.	T-076, T-077
ADR-022	Final re-review (T-079) dispositions: C1 drill 0.9 mm (F2-001); RV1 frame stays on AGND, plastic knob ≤ 15 mm on an insulated shaft and a knob/shaft strike in the REQ-023 test, risk R-011 (F2-002); REQ-032 evidence corrected (F2-003); assembly-guide wording (F2-004, F2-005, F2-007); spark-gap check in `pcb_rules_check` (F2-006); INTERFACES frozen-7.	T-080, T-081
ADR-023	CR-003 SMD conversion: one copper layer on F.Cu, SMD only, 4 M3 holes; Pico on castellations, module on butted pins, RV1 off-board, J2 solder-wire pads; bare MPR121 rejected.	T-090, user
ADR-024	BOM moved to Mouser (R1–R12 choice superseded by ADR-025 and ADR-026).	T-095, user

ADR	Decision	Basis / evidence
ADR-025	R1–R12 HPWC2512 3k3 (superseded by ADR-026).	User
ADR-026	R1–R12 YAGEO RC2512FK-073K3L 3k3 1 %; pulse capability NOT PROVEN, accepted by the user.	User
ADR-027	REQ-032 basis ex shipping (PASS); plan B trigger (R-018); R-016 stock risk accepted; INTERFACES frozen-8; JP1 etched legend accepted (REQ-040 FAIL).	Manager, T-099
ADR-028	CR-003 review dispositions: Pico 0.11 mm short of flush; ORIG label text corrected; PCB for Pico and Pico 2 only; BOM consistency; R-016 alternative; residual items for the first build; frozen-9 after T-105 and the re-review (Section 3.5).	Manager, T-103, T-104

Change requests. The candidate CR-001 (pulse-rated resistor, not opened) is superseded by CR-002 [`changes/CR-002.md`], which handles the pulse rating together with the flashover gap across the resistor (ADR-019, Section 6.5). The Vishay MBB 0207 named in CR-001 is rated only 1.74 kV at 3.3 kΩ (SOURCE EE-S17b), below the 3.64 kV needed (CALCULATION).

5 Interfaces

v1.3 note. The current revision is `frozen-8` (CR-003, frozen by the manager, ADR-027 D-4), with a `frozen-9` proposal for the T-103 board (A2 lands, module height, Pico mounting and keep-outs, 1206 lands, J1 out of the BOM) to be frozen after documentation T-105 and the CR-003 re-review (ADR-028 D-8) [`management/INTERFACES.json`]. Nets, pin map, I^2C, USB MIDI and build flags are unchanged in v1.3. The text below describes the v1.2 revisions.

Source: [`management/INTERFACES.json`], revision `frozen-7` (CR-002 final re-review follow-up, FrutaMIDI v1.2, T-080/T-081, ADR-022; frozen by the manager on 2026-11-08). History: frozen-1 of 2026-11-03 (ADR-013); documentation-only changes from T-012 (ADR-016) and review N-003 (ADR-018) gave frozen-3 of 2026-11-04. Frozen-4 changes the R1–R12 part and footprint, the IF-TERMINALS protection text, the board revision label, and records the v1.2 placement for traceability; nets, pinout, terminal map, I^2C, USB MIDI and build flags are unchanged. Frozen-5 (T-074) refreshed the evidence notes and the H3 placement after the re-review. Frozen-6 (T-077) adds C1 (100 nF, pot wiper to AGND), puts the RV1 mounting tabs on AGND (AGND is now exactly A1 pin 33, RV1 pin 3, the RV1 tabs and C1 pin 2), and records the JP1 footprint (3.81 mm pitch, 1.87 mm pad gap) and the 3.0×4.5 mm oval J2 pads; pinout, terminal map, I^2C, USB MIDI and build flags are unchanged. Frozen-7 (T-081, ADR-022) is a documentation-only change of the mechanical record: C1 drill $0.8 \rightarrow 0.9$ mm (annular ring 0.55 mm, finding F2-001), the $Ex \leftrightarrow$ GND edge-track spark gap of 1.40 ± 0.05 mm now enforced by `pcb_rules_check.py` (F2-006), and an AGND ESD note for the RV1 knob/shaft path (risk R-011, F2-002); pinout, nets, terminal map, I^2C, USB MIDI and build flags are unchanged. Changes go through the engineering change process.

Interface	Summary	Requirements
IF-USB-MIDI	Full-speed USB, Audio/MIDIStreaming USB MIDI 1.0, bulk IN/OUT 64-byte packets; bus powered, bMaxPower 100 mA (SOURCE S-11); product string “FrutaMIDI”. Sends Note On (0x9n), Note Off (0x8n) and Polyphonic Key Pressure on channel `MIDI_CHANNEL` (default 1, range 1–16); incoming MIDI is read and discarded. Hosts: Windows 10/11, macOS 12 or later.	REQ-011, 012, 013, 019

Interface	Summary	Requirements
IF-I2C	400 kHz; sensor 0 at 0x5A, optional sensor 1 at 0x5B (breadboard). PCB pull-ups R13 (SCL), R14 (SDA) 4.7 kΩ to +3V3, plus module 10 kΩ pull-ups and BSS138 shifters. 3.3 V logic. Series R15 (SCL) and R16 (SDA) 220 Ω on the Pico side; fault current 15.5 mA per line (CALCULATION at 3.4 V, ≤ 50 mA IIOVDD, SOURCE S-17).	REQ-021, 025, 026, 031
IF-PICO-PINS	Breadboard: SDA GP4 (pin 6), SCL GP5 (pin 7), button GP15 (pin 20), pot GP26/ADC0 (pin 31, optional), 3V3 pin 36. PCB: SDA_PICO GP26 (pin 31) via R16, SCL_PICO GP27 (pin 32) via R15, button GP22 (pin 29), pot GP28. AGND (pin 33) is the analogue ground return (SOURCE S-01 §2.1); on the PCB it reaches only RV1 pin 3, the RV1 mounting tabs and C1 pin 2 (ADR-021). LED GP25 (Pico, Pico 2) or CYW43 WL_GPIO0 (W boards) (SOURCE S-02).	REQ-017, 031
IF-TERMINALS	J2: 14 terminals at 5.08 mm: GND, E0–E11, GND; pads 3.0 × 4.5 mm oval, drill 1.4 mm, copper extended 0.75 mm towards the back of the block (ADR-021). Terminal k ($k = 2..13$) reaches MPR121 ELE($k-2$) through R1–R12 3.3 kΩ, Vishay MBE04140C3301FC100 (DIN0414, lying flat, 15.24 mm lead pitch); protection at ±4 kV is an estimate on ASSUMPTIONs, not a pass (review F-002); resistor pulse rating FAIL at 8 kV (R-006).	REQ-022, 023, 030
IF-MPR121-MODULE	Socket A2: 1×13 electrode row and 1×7 power/I²C row S1–S7. JP1 (upright since v1.2: QT top, ORIG bottom; 3.81 mm pitch, 1.87 mm pad gap since ADR-021): pad 1 S1 (ORIG), pad 2 +3V3, pad 3 S7 (QT); a wrong choice puts +3V3 on ADDR or IRQ, never GND (REQ-039 PASS, REQ-040 FAIL). J1: pad 1 SCL, pad 2 SDA; wires to S5/S6 (original) or S3/S4 (STEMMA QT).	REQ-024, 025
IF-CONTROLS	SW1 to GND with internal pull-up, sampled every 10 ms; long press > 800 ms recalibrates, short press selects the next scale. RV1 10 kΩ linear (plastic knob ≤ 15 mm on an insulated shaft, R-011), wiper filtered by C1 100 nF to AGND ($\tau \leq$ 0.38 ms, CALCULATION): 10-bit ADC read every 30 ms, 5 zones of 0.66 V, hysteresis 0.7 zone, dead band 0.132 V against an error budget of 0.0364 V (CALCULATION, budgets.md).	REQ-006, 007, 008
IF-FW-BUILD	arduino-cli 1.5.1, arduino-pico 6.3.1, Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2. FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for Pico, Pico W, Pico 2, Pico 2 W; `-DFRUTAMIDI_PCB` selects the PCB pin map; optional `arduino:avr:leonardo`. Artifacts `FrutaMIDI-<board>.uf2` and `FrutaMIDI-pcb-<board>.uf2`.	REQ-033
IF-PCB-AUX	Documentation-only (review N-003): J3 = the S1–S7 wire holes (FrutaMIDI:WireHoles_1x07_P2.54mm), each joined to the module power/I²C row; the female-header sockets (2×1×20 Pico, 1×13 and 1×7 module) are a Socket field of A1/A2, not separate symbols.	—

Board: outline 100 × 70 mm, copper on B.Cu only, label "FrutaMIDI v1.2" (ADR-019 D-2; text in `hardware/scripts/gen_pcb.py` and `hardware/frutamidi.kicad_pcb`), fabrication folder `hardware/fabrication`. The placement recorded in frozen-6 and frozen-7 (for example A2 at (60.02, 17.645) mm, H4 at (88.5, 30.5) mm, C1 at (31.2, 5.3) mm, MEASUREMENT on the design file) is for traceability and is not a frozen mechanical interface. Mounting hole H3 moved from (95, 48) to (96, 38) mm in T-073 (review finding C2-009); the placement record in `management/INTERFACES.json` was updated by systems in T-074, and C1 was added in T-077.

6 Electronics analysis

v1.3 note. This section is the electronics analysis of the v1.2 through-hole board (pulse-rated MBE0414 resistors, screw terminals, on-board potentiometer, 133 holes). For the v1.3 SMD board, its parts, cost and ESD analysis see Section 3; the I^2C and sensitivity reasoning below still applies with the v1.3 values given there.

Sources: [`reports/electronics/analysis.md`] (T-050/T-052 and, for CR-002, §5 by T-060 and T-073 and §6 by T-076 and T-080), handoffs T-053, T-054, T-055, T-060, T-073, T-076 and T-080 in [`management/handoffs/electronics-engineer/`], raw outputs in [`sim/electronics/calc/`]. Nothing in this section is a measurement on hardware; "MEASUREMENT (design file)" means a dimension read from the KiCad file by a script.

6.1 Schematic and PCB checks

The schematic and PCB are generated by `hardware/scripts/` and checked by `build_all.sh` with kicad-cli 10.0.4. For v1.2 after the final re-review fixes (T-080 electronics and T-081 systems, committed in a51dfff; before that the manager PCB review follow-up T-076/T-077, committed in a216c1b and f457bfe) [`hardware/build/erc.rpt`] [`hardware/build/drc.rpt`] [`reports/electronics/evidence/T-080/`]:

- ERC: "ERC messages: 0 Errors 0 Warnings 0".
- DRC with schematic parity: "Found 0 DRC violations", "Found 0 unconnected pads", "Found 0 Footprint errors".
- Home-etch rules (`sim/systems/pcb_rules_check.py`): "SUMMARY: 20 checks, 20 PASS, 0 FAIL", including "ADR-021 JP1 pad gap >= 1.8 mm" (gaps 1.87/1.87 mm), "ADR-021 J2 pads 3.0 x 4.5 mm" (14 pads), the C1 annular ring ("min ring 0.55 mm ... drill 0.9") and, new in T-081 (finding F2-006), "ADR-020 Ex <-> GND edge-track spark gap 1.40 +/- 0.05 mm" ("gap min 1.400 / max 1.400 mm", 12 of 12 pads) [`sim/systems/pcb_rules_check.out`]. Mutation tests on a scratch copy make the spark-gap check FAIL as intended: "gap min 0.650" with the J2 drill offset removed, "gap min -0.100" with the offset sign flipped [`management/handoffs/systems-engineer/T-081/`].
- Interface check: "SUMMARY: 85 checks, 85 PASS, 0 FAIL", including `pcb.AGND_membership` (AGND pads A1/33, C1/2, RV1/3, RV1/MP) [`sim/systems/interface_check.out`] (re-run in T-080 and T-081). The intermediate T-076 run gave 79/80 with the expected FAIL of the old rule `pcb.A1.AGND_isolated`, which T-077 replaced.
- Silkscreen (`check_silk.py`, now also footprint designators and text-to-text): "PASS worst silkscreen-text to pad gap 0.35 mm ('R13' <-> R13 pad 1), 34 texts" and "PASS worst silkscreen text-to-text gap 0.11 mm" (MEASUREMENT, design file) [`reports/electronics/evidence/T-076/check_silk.out`].

These are design-rule results, not hardware validation.

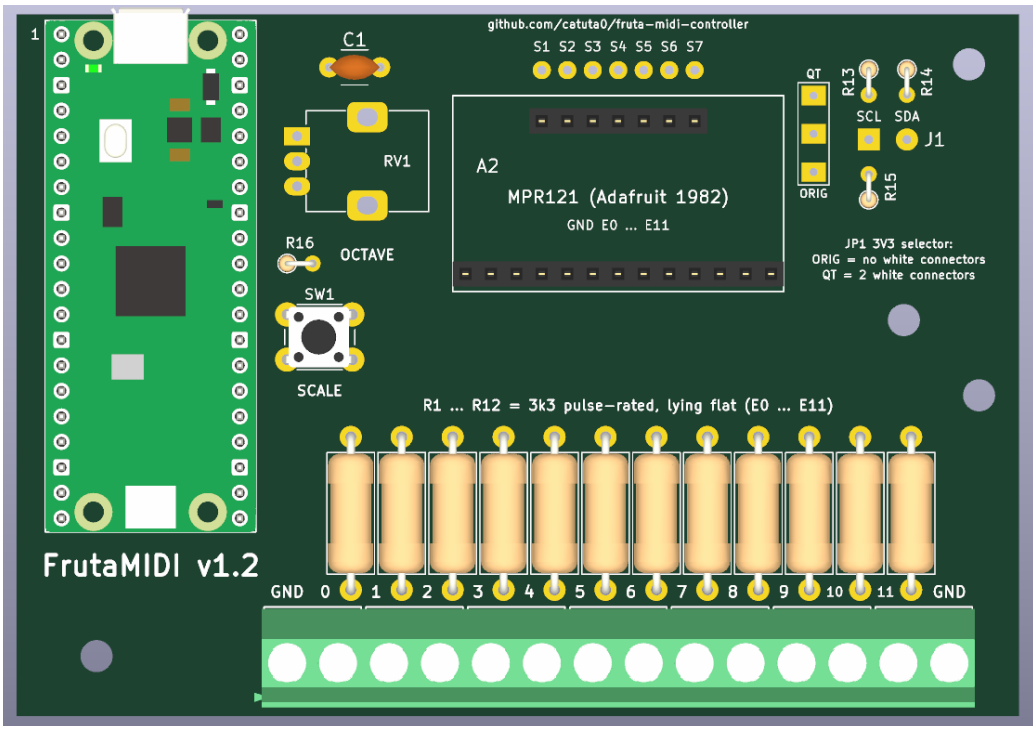


Figure 2: FrutaMIDI v1.2, component side (KiCad 3D render). R1–R12 lie flat in one row above the screw terminals; JP1 stands upright (QT top, ORIG bottom) to the right of the module on a 3.81 mm pitch (ADR-021), with J1 and R13–R15 next to it. C1 (100 nF) sits just above the octave potentiometer RV1. Mounting hole H3 sits at the right edge above R12, at (96, 38) mm since T-073. Render at a51dfff (T-080 regeneration; only the C1 drill and 3D model scale changed since a216c1b). [`hardware/fabrication/images/board-top.png`]

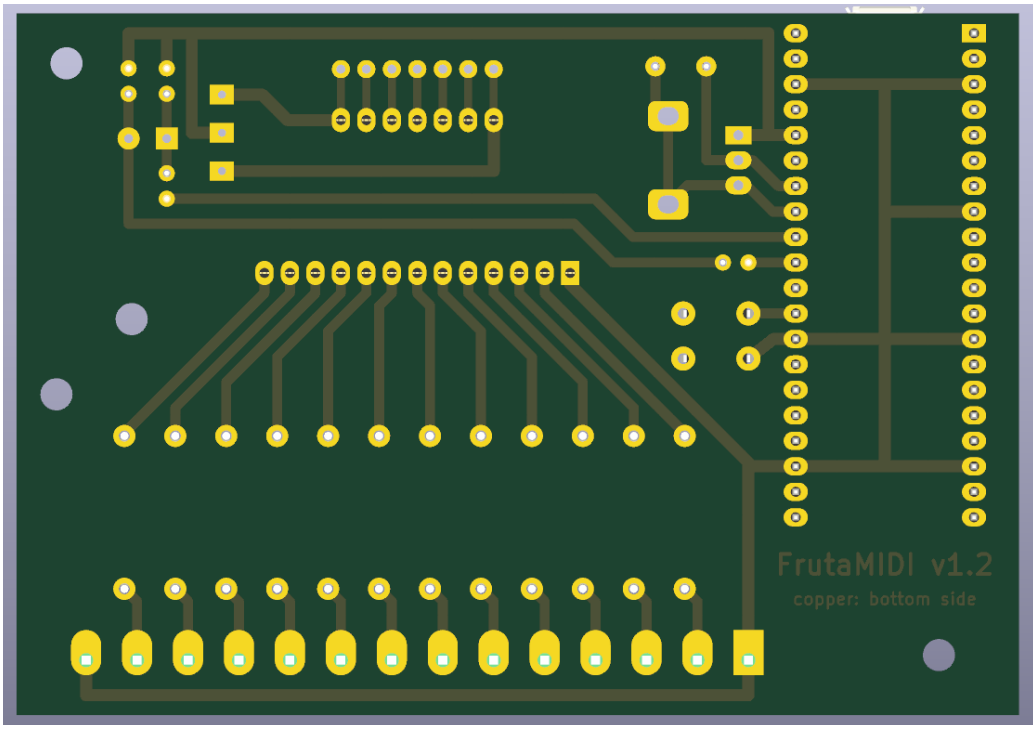


Figure 3: FrutaMIDI v1.2, copper side (bottom view). The ELE fan-out runs at 45° above the resistor row; the E side runs straight down to J2, whose pads are 3.0×4.5 mm ovals extended towards the back of the block (ADR-021). Render at a51dfff. [`hardware/fabrication/images/board-copper.png`]

6.2 Datasheet checks and budgets

- Pico pins (GP26/GP27 I^2C1 on pins 31/32, GP22 pin 29, GP28 ADC2 pin 34) match the firmware PCB map: PASS (SOURCE EE-S08).
- MPR121 VDD through the module regulator: AP2112 ≈3.295 V typical, ≥3.19 V worst; MIC5225 ≈3.25 V typical, ≥3.10 V worst; inside 2.0–3.6 V: PASS (CALCULATION).
- 3V3 load: 4.45 mA worst case, board power 14.7 mW (CALCULATION, electronics). Systems gives 10.15 mA worst case with 2 sensors and PCB pull-ups, 96.6 % margin to 300 mA (CALCULATION) [`reports/systems/budgets.md`] [`management/REQUIREMENTS.md`] (REQ-020).
- VBUS current worst case 55.6 / 70.6 / 60.6 / 75.6 mA for Pico / Pico W / Pico 2 / Pico 2 W against 100 mA (CALCULATION; upper bounds ASSUMPTION) [`reports/systems/budgets.md`].

6.3 Bill of materials and cost

Part numbers from [`hardware/fabrication/bom-kicad.csv`] (ADR-015, ADR-019): A1 Raspberry Pi SC0915; A2 Adafruit 1982; **R1–R12 Vishay MBE04140C3301FC100** (3k3 1 %, DIN0414, pulse-rated; was Yageo MFR-12FTF52-3K3 up to v1.1); R13/R14 MFR-12FTF52-4K7; R15/R16 MFR-12FTF52-220R; J2 Phoenix Contact 1715721 (×7); RV1 Alps Alpine RK09K series, 10 kΩ linear, vertical (order code not sourced); **C1 KEMET C322C104K5R5TA** (100 nF X7R 50 V ±10 %, radial, 5.08 mm leads; new in ADR-021; not the C320 version, which has 2.54 mm leads, SOURCE EE-S29); RV1 knob: plastic, ≤ 15 mm diameter, covering the whole shaft, on an insulated shaft where an RK09K variant has one (not confirmed from an Alps datasheet; ADR-022, risk R-011); SW1 Omron B3F-1000; female headers generic (a Socket field of A1/A2). JP1 is copper only, and the J2 pad change costs

nothing [management/handoffs/electronics-engineer/T-076.md].

Item	v1.1 (budgets rev. 5)	v1.2 (budgets rev. 7)	Label
R1–R12, 12 parts	0.24–0.60 EUR	3.60–7.92 EUR	v1.2: SOURCE EE-S26 sibling values, equal 3k3 price ASSUMPTION; v1.1: ASSUMPTION
C1 100 nF (ADR-021)	—	0.20–0.90 EUR	ASSUMPTION from a distributor search, EE-S30, not verified
Core BOM total	20.42–30.40 EUR	23.98–38.62 EUR	CALCULATION
Over the 30 EUR target (upper bound)	0.40 EUR	8.62 EUR	CALCULATION
REQ-032 status	FAIL	FAIL	—

Sources: [reports/systems/budgets.md] [management/REQUIREMENTS.md] (REQ-032) [management/metrics.json] (bom_core_range, pot_filter_C1_price). The R1–R12 increase is accepted for ESD safety (ADR-019 D-3) and the C1 increase by the ADR-021 addendum; the requirement stays FAIL and is not waived. Revision 6 (before C1) gave 23.78–37.72 EUR. Most other prices are ASSUMPTIONs; a quote from one supplier would narrow the range.

6.4 Electrode path (critical block)

Model sim_electrode.py (SIMULATION): constant-current charge into a 17 pF pin plus series R plus external C, swept over R = 0–100 kΩ and C = 10–2000 pF; the discharge model is an ASSUMPTION [reports/electronics/analysis.md].

Variant	Peak 4 kV	Peak 8 kV	Sensitivity	Verdict
a) none	15.0 A	30.0 A	1.000	FAIL (11× over)
b) 3.3 kΩ series (selected)	1.10 A	2.20 A	0.904	Estimate at 4 kV (review F-002), PARTIAL 8 kV
b') 10 kΩ series	0.39 A	0.77 A	0.679	FAIL (sensitivity)
c) 3.3 kΩ + TPD4E05U06	1.10 A	2.20 A	≈0.90	FAIL (SMD constraint)
d) 3.3 kΩ + 2×1N4148	1.10 A	2.20 A	≈0.90	PARTIAL (12 jumpers)

- ESD model: IEC 150 pF/330 Ω; pin limit 1.33 A inferred from HBM 2 kV (ASSUMPTION). Minimum R: 2.67 kΩ at 4 kV, 5.67 kΩ at 8 kV. Design rule 2.7–3.3 kΩ; chosen 3.3 kΩ. The v1.2 part keeps the value and the 1 % tolerance, so this sensitivity analysis stays valid.
- With 3.3 kΩ the smallest detectable touch is 0.5 pF on 10 pF and 19.9 pF on 1000 pF (19.8 pF without R) (SIMULATION).
- The breadboard build has no electrode protection beyond the MPR121's internal structures (HBM ±2 kV, SOURCE EE-S01), because the module wires ELE0–11 straight to its header (T-055).

6.5 ESD flashover around the series resistor and pulse rating (CR-002)

Sources: [reports/electronics/analysis.md] §5, [management/handoffs/electronics-engineer/T-060.md], [management/handoffs/electronics-engineer/T-073.md], raw outputs calc_flashover.out, calc_pulse_rated.out, check_gap.out, check_gap_old_86866be.out, check_gap_old_1247c44.out, calc_gnd_bounce.out and calc_mechanical.out in [sim/electronics/calc/]. Change request: [changes/CR-002.md]; decisions ADR-019 and, after the re-review (Section 9.1), ADR-020.

Mechanism. During a contact discharge almost the whole IEC voltage appears across the series resistor, because the pin structure holds the ELE side near 0 V (ASSUMPTION): $V_R = V \cdot R/(R + 330\,\Omega) = 3636$ V at 4 kV and 7273 V at 8 kV (CALCULATION). Two things can then defeat the resistor: an arc across the air gap between its two ends, which leaves the pin with only the 330 Ω source (12.1 A at 4 kV and 24.2 A at 8 kV, against 1.10 A and 2.20 A through R, CALCULATION); or a resistor body not rated for that voltage.

Correction of an earlier claim. The v1.1 analysis stated that "each path still has its series R". That was wrong for the gap across the resistor itself: with the vertical DIN0204 footprint (pads 1.6 mm on 2.54 mm pitch) the ELEx–Ex copper gap was 0.94 mm (MEASUREMENT, design file of 86866be), and the vertical lead ran about 1 mm from the opposite end cap (ASSUMPTION). At an assumed 3 kV/mm breakdown that gap flashes over at about 2.8 kV, below the 3.64 kV across R (CALCULATION, CR-002). The resistor (Yageo MFR-12, 400 V maximum overload, SOURCE EE-S14b) was also not rated for the pulse.

Breakdown sweep and gap rule. Every breakdown model is an ASSUMPTION; the constants are stated in calc_flashover.py. The IEC 60664-1 values were recalled, not re-read (EE-S25), and must be checked against the standard.

Model	Gap at 4 kV (3.64 kV)	Gap at 8 kV (7.27 kV)
Uniform field 4.5 / 3.0 / 2.0 kV/mm	0.81 / 1.21 / 1.82 mm	1.62 / 2.42 / 3.64 mm
Paschen (Townsend form, $\gamma = 0.01$)	0.67 mm	1.56 mm
Empirical uniform air (Kuffel)	0.73 mm	1.79 mm
Non-uniform, $\eta = 0.7$ / 0.5 / 0.3	1.17 / 1.79 / 3.33 mm	2.77 / 4.13 / 7.46 mm
IEC 60664-1 Table F.2 case A, interpolated (recalled)	2.61 mm (3.00 mm at the full 4 kV)	7.07 mm

Design rule (CALCULATION, worst of $\eta = 0.3$ and F.2 case A, rounded up): (a) ELE$x \leftrightarrow$ Ex copper and leads of one channel and (b) E$x \leftrightarrow$ any other channel's ELE net or any MPR121/Pico net $\geq$ **3.5 mm** for 4 kV; 8 kV would need $\geq$ **7.5 mm**. E$x \leftrightarrow$ Ey and E$x \leftrightarrow$ GND are only reported: an arc there still passes a resistor, or diverts the strike to ground.

Part selection. The resistor body must also hold V_R. The 1.2/50 µs ratings at 3.3 kΩ read from the vector chart of Vishay document 28766 (SOURCE EE-S17b; ±5 % reading ASSUMPTION):

Candidate (3.3 kΩ)	Rating	Against 3.64 kV (4 kV case)
Yageo MFR-12 (v1.1 part)	400 V max. overload (EE-S14b)	FAIL
Vishay MBA 0204	1.47 kV	FAIL
Vishay MBB 0207	1.74 kV	FAIL
Vishay MBE 0414, MBE04140C3301FC100 (selected)	**4.08 kV**	PASS, margin **1.12** (1.07 with −5 % reading) (CALCULATION)
Vishay LSR37	12 kV ESD contact (EE-S24)	PASS at 8 kV too, but ±10 % and up to ±20 % drift after 100 strikes

Ratings are SOURCE; the margin is rating/V_R = 4.08/3.64 (CALCULATION).

Pulse energy (corrected in T-073, finding C2-003). The 1.2/50 µs rating pulse puts $(V_{pk}^2/R)\int(V/V_{pk})^2\,dt = 4083^2/3300 \cdot 36.06\,\mu s \approx$ **182 mJ** into 3.3 kΩ (CALCULATION; "Rating pulse energy: integral (V/Vpk)^2 dt = 36.06 us ... -> 182 mJ into 3k3 at 4083 V"). The T-060 value of about 364 mJ used $\int V\,dt$ and was about twice too high. Below its knee the MBE 0414 chart has a log-log slope of 0.496, i.e. constant V^2/R: at 3.3 kΩ the rating is **energy-limited**; above the knee at **5073 Ω** it is voltage-limited at a plateau of **5054 V** (CALCULATION from the SOURCE chart). One 4 kV IEC strike puts 1.09 mJ and 3636 V into R: 167× below the rating-pulse energy and 1.39× below the plateau; at 8 kV, 4.36 mJ and 7273 V exceed both the 3.3 kΩ chart value and the plateau (CALCULATION). The 3.3 kΩ chart value 4083 V stays the acceptance limit (conservative); verdicts unchanged: PASS at 4 kV, FAIL at 8 kV. Taking the 1.2/50 µs rating as a proxy for the ≈0.54 µs IEC pulse is an ASSUMPTION. The MBE 0414 keeps the 3.3 kΩ 1 % value, so the sensitivity analysis above is unchanged (ADR-019 D-1).

Layout (v1.2). R1–R12 lie flat in one row with their axis along y, 15.24 mm lead pitch (datasheet minimum 15.0 mm, SOURCE EE-S17b), pads 2.2 mm, drill 1.1 mm since T-073 (was 1.0 mm; finding C2-008): annular ring (2.2 − 1.1)/2 = 0.55 mm ≥ 0.40 mm and hole − lead = 0.30 mm diametral ≥ 0.2 mm for the 0.8 mm lead (CALCULATION; `calc_mechanical.out` "R_Axial_DIN0414... 1.10 2.20 24 0.55 0.80 0.30 PASS"). The ELE fan-out runs at 45° above the row and the E side runs straight to J2 (Figures 2 and 3). To make room, the MPR121 module moved 15.1 mm up and 2.54 mm right; J1, JP1 (now upright) and R13–R15 moved to the right of the module; H4 moved from (70, 6) to (88.5, 30.5) mm; in T-073 H3 moved from (95, 48) to (96, 38) mm (see "Mounting metal" below). Pinout, nets and the J2 row are unchanged (ADR-019 D-4).

Gaps in the design file. MEASUREMENT on `hardware/frutamidi.kicad_pcb` with `check_gap.py` (KiCad 10.0.4 Python):

Gap	v1.1 (86866be)	v1.2	Against 3.5 mm (4 kV) / 7.5 mm (8 kV)
(a) ELEx ↔ Ex copper, same channel	0.94 mm	13.04 mm (all 12)	v1.1 FAIL; v1.2 PASS / PASS
(b) Ex ↔ other ELE or MPR121/Pico net	1.99 mm	10.92 mm	v1.1 FAIL; v1.2 PASS / PASS
Ex ↔ Ey (reported)	0.80 mm	2.08 mm	reported only
Ex ↔ GND (reported)	1.40 mm	1.40 mm	reported only; intentional spark gap (below)

Proof lines (T-073 re-run, c901ad8): "PASS (a) copper gap 13.04 mm vs rule 7.5 mm (8 kV)", "PASS (b) copper gap 10.92 mm vs rule 7.5 mm (8 kV)"; v1.1 "SUMMARY: FAIL" [`sim/electronics/calc/check_gap.out`] [`sim/electronics/calc/check_gap_old_86866be.out`]. These are copper gaps only. The T-060 "lead ↔ lead 14.44 mm" line was removed in T-073 (finding C2-002): the leads of a flat resistor leave the end caps outwards and are never the closest metal. Copper values are also recorded in [`management/metrics.json`] (`esd_gap_ele_e_min`, `esd_gap_e_other_min`, `electrode_R_pulse_rating_3k3`).

End caps on the component side (T-073, finding C2-002). The closest metal of a flat resistor is cap to cap, not copper to copper. The Vishay drawing gives no cap length, so it is swept: cap length L_c 1.5–3.5 mm (ASSUMPTION), body length 11.9 mm (SOURCE, $L_{\max}$) and 11.0 mm (ASSUMPTION, no $L_{\min}$ given), cap diameter 4.2 mm (SOURCE), lacquer ignored, axial placement error of a hand-fitted body 0 / ±0.25 / ±0.5 mm (ASSUMPTION). Same channel the gap is $L - 2L_c$; the cross-channel pair (E cap of Rk to ELE cap of R$k \pm 1$) is the worst (CALCULATION, `check_gap.out` section (d)):

L / L_c (mm)	Same channel	Cross-channel, no error	±0.25 mm	±0.5 mm
11.9 / 1.5	8.90	8.94	8.45	7.95
11.9 / 2.5	6.90	6.96	6.46	5.97
11.0 / 2.5	6.00	6.06	5.57	5.08
11.0 / 3.0	5.00	5.08	4.59	4.10
11.0 / 3.5 (worst)	4.00	4.10	3.61	**3.13**

Values in mm, CALCULATION on the ASSUMPTIONs above (full sweep in [`sim/electronics/calc/check_gap.out`]). Proof lines: "PASS (d) caps worst 3.61 mm vs rule 3.5 mm (4 kV); FAIL vs 7.5 mm (8 kV) [placement error +-0.25 mm ...]" and "FAIL (d) caps worst 3.13 mm vs rule 3.5 mm (4 kV) ... [placement error +-0.50 mm ...]". Result: at 4 kV **PASS** over the whole sweep for placement error ≤ ±0.25 mm (worst **3.61 mm**); **FAIL** only at the corner L_c 3.5 mm, L 11.0 mm with ±0.5 mm (**3.13 mm**); for $L_c \leq 3.0$ mm it passes at every swept L and error. Status **PARTIAL** (rests on an assumed cap length). In short: about 4.0 mm with centred bodies ("PASS (d) caps worst 4.00 mm ... [placement error +-0.00 mm" over the sweep), at least 3.6 mm within ±0.25 mm, about 3.1 mm at ±0.5 mm. Mitigation (ADR-020; wording of the assembly guide corrected after finding F2-004, T-080): the assembly guide gives these three values and requires the leads to be formed on a lead-forming gauge (or a stripboard strip), not by eye, so the two straight lengths are equal within about 0.25 mm, and the first-article check measures L and L_c with a caliper (V-26). At 8 kV the 7.5 mm rule is met only by copper; the end caps FAIL ("SUMMARY 8 kV: copper PASS; end caps FAIL (worst 3.13 mm vs 7.5 mm)").

Mounting metal (T-073, finding C2-009). A metal screw head, washer or spacer (radius 2.8–3.5 mm, ASSUMPTION) at a mounting hole can bridge two caps; an arc ELEx → metal → Ex is counted as the sum of the two gaps (ASSUMPTION). At the old H3 (95, 48) mm the bypass sum was 1.94 mm: FAIL against 3.5 mm ([`sim/electronics/calc/check_gap_old_1247c44.out`]). H3 moved to (96, 38) mm, where it faces only the ELE11 cap: worst bypass over H1–H4 "PASS (e) floating-metal bypass worst 15.36 mm vs rule 3.5 mm (4 kV); PASS vs 7.5 mm (8 kV)" (CALCULATION). Nylon hardware at H3 is still recommended in the assembly guide.

Ex ↔ GND 1.40 mm kept as an intentional spark gap (T-073, finding C2-005, ADR-020). Every J2 electrode pad is 1.40 mm (MEASUREMENT, design file) from the GND edge track, below the breakdown distance of every model at 3.64 kV (0.67–3.33 mm), so a 4 kV strike

most likely arcs to GND there. That keeps the strike off Rx and the MPR121 pin; moving the track away would put the full 1.10 A into the pin through Rx. The arc current then raises the board ground. GND track lengths (MEASUREMENT, design file): terminal GND to Pico pin 23 24.04 mm, module GND pin to Pico pin 23 31.36 mm, shared section 4.72 mm. With the IEC 4 kV current (SOURCE EE-S16, recalled from memory: first peak 17.2 A at the upper tolerance, rise 0.6–1.0 ns), track inductance 0.7–1.0 nH/mm and socket plus header pin 3–10 nH (ASSUMPTION), the module-GND to Pico-GND bounce is **87–339 V for ≈1 ns**, about 4.5 V after the first peak; through R15/R16 (220 Ω, then the module's BSS138 level shifters) that is **≤1.53 A** and ≤0.92 nC per I^2C line (CALCULATION on ASSUMPTIONs; "Worst case: module GND - Pico GND bounce 339 V for ~1 ns", [`sim/electronics/calc/calc_gnd_bounce.out`]). The RP2040 HBM 4 kV rating (SOURCE EE-S28) is stated for digital FT pins; no HBM value was found for GP26/GP27, which carry I^2C1 and are ADC-capable (NOT CONFIRMED, escalated). Residual risk: an I^2C pin upset or latch-up from the ≈1 ns bounce. It is an estimate, not a measurement; the ESD test watches for an arc to the GND track and for I^2C errors and resets after each strike (V-10, V-25). Since T-081 (finding F2-006) the gap is enforced by `pcb_rules_check.py` at 1.40 ± 0.05 mm (12 of 12 pads at 1.400 mm, MEASUREMENT, design file), so a flipped or removed J2 copper offset (0.65 mm, which still passes DRC) can no longer go unnoticed.

Knob and shaft discharge into AGND (finding F2-002, risk R-011). The RV1 mounting tabs and frame are on AGND (ADR-021), and on the RK09K a metal shaft is assumed to touch the frame (ASSUMPTION, datasheet not opened). A finger discharge to a metal shaft or a conductive knob would therefore enter AGND, which joins the Pico at pin 33, with no series resistor (the electrodes have 3.3 kΩ). This path is not analysed numerically: no peak current, no flashover path and no RP2040 ADC/AGND ESD rating (EE-S28 covers digital FT pins). Mitigation (ADR-022): a plastic knob, at most 15 mm in diameter, covering the whole shaft, on an insulated shaft where available; a knob/shaft strike is added to the REQ-023 physical test (step 2b, V-29). The fallbacks (tabs unconnected, or tied to GND at the board edge) apply only if that test fails. Risk R-011 is OPEN [`management/RISKS.md`].

8 kV deferred. At 8 kV the copper gaps meet the 7.5 mm rule but the end caps do not, and the selected resistor does not hold 7.27 kV (4.08 kV rating): FAIL, and the pin current of 2.20 A exceeds the inferred 1.33 A limit by 1.65× (CALCULATION on ASSUMPTION). The scope stays ±4 kV (REQ-023, ADR-019 D-1). The LSR37 (same DIN0414 footprint, 12 kV rated) is a later option; it is deferred because its tolerance and drift bring touch sensitivity to about 0.89–0.90 against the 0.90 criterion (T-060). Risk R-006 (accepted).

Status. Copper flashover around R: PASS on design data at the 4 kV and 8 kV rules. End caps: PARTIAL at 4 kV (assumed cap length and placement), 8 kV not shown. Mounting metal: PASS. Ex ↔ GND: intentional spark gap, enforced by a design-file check, residual risk estimated, not measured. RV1 knob/shaft path: not analysed (R-011), test planned (V-29). Resistor pulse rating: PASS at 4 kV, FAIL at 8 kV. REQ-023 stays PARTIAL until an ESD-gun test (V-10, V-11, V-25, V-26 and V-29 in Section 11).

6.6 I^2C bus

- At e4668d4 the rise time with only the module 10 kΩ pull-ups was 322 ns at 38 pF against 300 ns: FAIL (CALCULATION, bus C partly ASSUMPTION). With R13/R14 4.7 kΩ: 103 ns at 38 pF (electronics) and 156 ns at 60 pF (systems, RPU 80 kΩ SOURCE S-17): PASS (CALCULATION) [`reports/systems/budgets.md`].
- Series R15/R16 220 Ω: fault current 15.5 mA per line, 30.9 mA for both, against 50 mA (CALCULATION). Low-level margins (authoritative, RP2040 datasheet levels SOURCE

S-17): 0.138 V when the Pico drives low (VIL 0.93 V at the MPR121) and 0.235 V when the MPR121 drives low (VIL 0.8 V at the Pico): PASS (CALCULATION) [`reports/systems/budgets.md`]. The earlier electronics figures (0.218 V and 0.286 V, with RP2040 levels as ASSUMPTION, T-055) are superseded interim values.
- Breadboard without the 4.7 kΩ pull-ups: 452 ns at 60 pF, FAIL above 39.8 pF; hence ADR-009 D-11 (CALCULATION).
- v1.2 configuration wires J1 → S holes are about 25–35 mm long (T-060): about 1.8 pF per line at an assumed 50 pF/m (CALCULATION on ASSUMPTION), against the 115.2 pF bus limit at 400 kHz (CALCULATION) [`management/RISKS.md`] (R-008).

6.7 Octave potentiometer filter and AGND (ADR-021)

Sources: [`reports/electronics/analysis.md`] §6, [`management/handoffs/electronics-engineer/T-076.md`], [`sim/electronics/calc/calc_pot_rc.out`], [`sim/systems/budgets.out`] (check A7). C1, 100 nF X7R (KEMET C322C104K5R5TA), runs from the pot wiper (net POT, GP28) to AGND just above RV1; the RV1 mounting tabs are soldered and tied to AGND (symbol `Device:R_Potentiometer_MountingPin`). The AGND net is now exactly A1 pin 33, RV1 pin 3, the two RV1 tabs and C1 pin 2; the only AGND–GND join stays inside the Pico (SOURCE S-02), and the interface check enforces that no AGND part other than A1 has a GND pad (`pcb.AGND_membership`, `pcb.AGND_no_shared_pad`, `pcb.AGND_parts_not_on_GND`: PASS).

C1 holes (T-080, finding F2-001). The C322 lead diameter is 0.48–0.61 mm (SOURCE EE-S29, .020 +.004/−.001 in). The T-076 drill of 0.8 mm left 0.8 − 0.61 = 0.19 mm at the largest lead, below the 0.2 mm fit rule. T-080 drills C1 at **0.9 mm** with the 2.0 mm pad kept: hole − lead = 0.9 − 0.61 = **0.29 mm** diametral ≥ 0.2 mm, annular ring (2.0 − 0.9)/2 = **0.55 mm** ≥ 0.40 mm, pad-to-pad gap 5.08 − 2.0 = 3.08 mm (CALCULATION; `calc_mechanical.out` "C_Radial_KEMET_C322_P5.08mm_HomeEtch 0.90 2.00 2 0.55 0.61 0.29 PASS"). A substitute capacitor fits if its leads are at most 0.7 mm (0.9 − 0.7 = 0.2 mm, CALCULATION) and its body is at most 5.08 × 3.18 mm and 6.6 mm tall. Drill file after T-080 (MEASUREMENT of the export `hardware/fabrication/gerbers/frutamidi.drl`): 0.8 mm 11, 0.9 mm 11, 1.0 mm 63, 1.1 mm 28, 1.4 mm 14, 1.8 mm 2 (slots), 3.2 mm 4; **133 holes** in total.

Knob clearance over C1 (finding F2-005). The RV1 shaft centre is 7.77 mm from the nearest edge of the C1 body, which may be up to 6.6 mm tall (CALCULATION from MEASUREMENT of the design file, T-080). A knob of at most 15 mm diameter (radius 7.5 mm) leaves 0.27 mm in plan, with no allowance for placement tolerance (ASSUMPTION that this is enough; checked at first build, V-28).

ADC settling. By CALCULATION: the wiper source resistance is $R\,x(1-x)$, largest at mid travel: 2.5 kΩ nominal, 3.0 kΩ worst (10 kΩ +20 %, ASSUMPTION). With C1 at 100 nF nominal and 126 nF worst (+10 % tolerance, +15 % X7R temperature, SOURCE EE-S29):

- $\tau = 0.25$ ms nominal (f_c 637 Hz), **0.38 ms** worst (f_c 419 Hz).
- A full-scale step settles to $\frac{1}{2}$ LSB at 10 bits in $7.62\,\tau = 1.91$ ms nominal, 2.89 ms worst, against the read period `POT_READ_MS` = 30 ms (SOURCE `firmware/FrutaMIDI/fm_pot.h`): margin **10.4**× worst (criterion ≥ 5×, ASSUMPTION).
- Lag during a 100 ms full turn (ASSUMPTION, fastest turn): 3.9 counts worst, against the ≈41-count hysteresis.
- Charge sharing with the RP2040 sample capacitor (≤ 10 pF, ASSUMPTION): ≤ 0.10 LSB.

Status: PASS by calculation ("PASS worst-case settling 2.89 ms vs read period 30 ms"); not measured. REQ-006 stays PARTIAL. The noise benefit of C1 has not been quantified; V-15 and V-28 in Section 11 measure it.

C2 (1 µF at the module) not added. The review proposed a 1 µF decoupling capacitor from +3V3 to GND near JP1. There is no GND copper in the right half of the board above the SDA run; on one layer without vias the only route would pass 3V3 within about 0.05 mm of an assumed H2 washer and run GND about 48 mm under the module, 1.1 mm from all twelve ELE pins. The resulting loop of about 100 mm (≈100 nH at an ASSUMPTIONed 1 nH/mm) would decouple little, and both module revisions carry their own 3.3 V regulator and capacitors (SOURCE EE-S02/EE-S03). Recorded as not needed (ADR-021).

6.8 Safety statement

Each PCB electrode has a 3.3 kΩ pulse-rated series resistor (Vishay MBE04140C3301FC100) lying flat, with at least 13.04 mm between the copper on its two sides (MEASUREMENT, design file); on the component side the end-cap gap is estimated at 3.61 mm or more for a hand-fitted part within ±0.25 mm (CALCULATION on an ASSUMPTIONed cap length, T-073). The J2 electrode pads sit 1.40 mm from the GND edge track, an intentional spark gap that diverts a strike to ground (ADR-020). Its protection against IEC 61000-4-2 contact discharge at ±4 kV is an estimate resting on ASSUMPTIONs (the pin limit is inferred from HBM 2 kV; the breakdown models and the pulse-rating proxy are assumed), not a pass (review finding F-002 [`reports/review/72f67f0/findings.md`]). Only an ESD-gun test can close it. At 8 kV the resistor rating is exceeded (R-006). A discharge to the potentiometer shaft would reach AGND without a series resistor; this path is not analysed, and a plastic knob of at most 15 mm on an insulated shaft is specified instead (R-011, ADR-022). The design is not evaluated for touching while connected to mains-powered equipment (ADR-003, ADR-012, ADR-019).

7 Firmware design and tests

Sources: [`reports/firmware/coverage.md`] and the T-040 to T-043 handoffs in [`management/handoffs/firmware-engineer/`].

7.1 Design

- One sketch `firmware/FrutaMIDI/FrutaMIDI.ino` with host-testable header-only logic `fm_*.h`: scales (`fm_scales.h`, English names), button (`fm_button.h`), debounce, aftertouch, potentiometer, non-blocking LED blinker (`fm_blink.h`), MPR121 burst reads (`fm_mpr121.h`).
- Pads: Note On at velocity 100 on channel 1 by default; Note Off always carries the stored note; debounce is a 15 ms lockout per pad.
- Aftertouch: Polyphonic Key Pressure 0–127 from delta 12–120; one 0 is sent before Note Off and on a fall to the floor (T-041).
- Scales: 6, Drum Rack (36–47, octave ignored) default; octave $-2..+2$ by potentiometer with 0.7-zone hysteresis.
- Shared notes (T-044, ADR-017): a reference count per note; Note On when the first pad takes a note, Note Off only when the last pad releases it, aftertouch is the maximum of the holders.
- Sensor health (T-044): 3 consecutive failed transactions, or a changed ECR read-back (every 250 ms), mark a sensor lost; its pads are released, the LED shows the fast error blink, and the sensor is restarted after more than 2000 ms (SOURCE `fm_sensor_health.h`). `setClock(400000)` now follows every sensor `begin()` (F-006).
- Defect found (ADR-017): on an I^2C NACK the Adafruit library's `touched()` returns 0x0FFF, i.e. all 12 pads touched, so a lost sensor would have sent 12 Note Ons. The sketch now reads the status registers itself and checks the transaction result [`management/handoffs/firmware-engineer/T-044.md`].

- Button: long press > 800 ms recalibrates (releases all notes, restarts sensors, 3 blinks); short press selects the next scale.
- LED patterns are non-blocking: pads stay live while the LED blinks (T-041). `startSensors()` still blocks the loop during a user-requested recalibration (kept by decision; duration to be measured). The automatic restart of a lost sensor also blocks it for 16.8–22.1 ms (CALCULATION), at most once per ≥2 s retry; REQ-016 lists this fault-path exception (ADR-018, review N-001).
- MPR121 configuration: AUTOCONFIG with USL/TL/LSL = 200/180/130 at design VDD 3.2 V and SCTS = 0 (ADR-010, ADR-011, SOURCE MPR121 datasheet S-006); I^2C stays at 400 kHz; run-based burst reads (worst mask 42 bus bytes in 8 transactions, CALCULATION) [`reports/systems/budgets.md`].

7.2 Latency budget

Budgets revision 5 (T-013, includes the ECR health poll) [`reports/systems/budgets.md`] [`sim/systems/budgets.out`], all CALCULATION for $C \leq 100$ pF at 400 kHz: worst case 10.72 ms (1 sensor) and 12.35 ms (2 sensors) against ≤ 15 ms; jitter 6.72 ms and 8.35 ms against ≤ 10 ms: PASS. A re-touch of the same pad within the 15 ms lockout adds up to 15 ms, giving up to 25.72 ms (1 sensor); REQ-014 is stated for a pad idle for at least 15 ms. At 100 kHz the jitter (1 sensor) and both 2-sensor checks FAIL, so the bus stays at 400 kHz (ADR-008).

7.3 Build and tests

Task	Result line (MEASUREMENT)	Status	Evidence
T-040	Host tests "213 passed in 1.33s"; 14/14 mutants fail the suite; compile matrix 9/9	PASS	`reports/firmware/T-040/`
T-041	Host tests "267 passed in 1.79s"; compile matrix 9/9	PASS	`reports/firmware/T-041/`
T-042	Host tests "268 passed in 1.61s"; datasheet check BLOCKED, later closed by ADR-011	PARTIAL	`reports/firmware/T-042/`
T-043	Host tests "268 passed in 1.97s"; rpipico PCB "Sketch uses 90700 bytes (4%) of program storage space", "Global variables use 17632 bytes (6%)"	PASS	`reports/firmware/T-043/checks.log`
T-044	Host tests "295 passed in 1.40s"; mutation check 10/10 killed; 9/9 builds exit 0 (rpipico PCB "Sketch uses 91196 bytes (4%)")	PASS	`reports/firmware/T-044/`

A clean compile and host tests are not hardware validation. The host loop harness copies the call order of `loop()` and does not run the sketch itself; the timing granularity on target depends on the loop period, which has not been measured [`reports/firmware/coverage.md`].

8 Host software and tests

Source: `docs/index.html` (browser sampler, GitHub Pages), handoffs T-020 and T-021 in [`management/handoffs/software-engineer/`], raw outputs in [`reports/software/`].

8.1 Design

- Web MIDI input, omni channel; 12 pads mapped to notes 36–47 by default; Note On with velocity 0 is treated as Note Off (REQ-034).
- Learning a note already used by another pad swaps the two pads' notes; notes received before audio is enabled flash the pad and emphasise the enable-sound button (ADR-005). All UI strings are in English (`lang="en"`).

8.2 Tests

- T-020: `run_pytest tests/host` with Python 3.9.6, pytest 8.4.2, node v26.4.0, playwright-core 1.64.1, Google Chrome 156.0.8078.74: "37 passed in 69.20s" (MEASUREMENT) [`reports/software/T-020/pytest_tests_host_verbose.txt`].
- T-021: same tools, Chrome headless: "102 passed in 56.45s" (MEASUREMENT), including the ADR-005 behaviour tests, a rendered language test over 8 states and mutation checks [`reports/software/T-021/pytest_tests_host_verbose.txt`].

The tests use a stubbed Web MIDI API in headless Chrome. No real MIDI device, no audible output and no person looking at the page in a real browser have been tested (T-020, T-021).

9 Independent review findings

v1.3 note. The CR-003 independent review (T-102) of the SMD board and its dispositions are in Section 3.5. The reviews below concern the v1.1 and v1.2 boards.

An independent reviewer, who did not produce the work, reviewed revision `72f67f0` (task T-060) [`reports/review/72f67f0/findings.md`] and then re-reviewed the fixes in `72f67f0..05a06a3` (task T-061) [`reports/review/05a06a3/findings.md`]. Dispositions are recorded in ADR-014 and ADR-018 [`management/DECISIONS.md`].

Summary.

- First review: 16 findings (0 critical, 3 major, 9 minor, 4 info); dispositions 13 FIX, 1 ACCEPTED, 2 info FIX (ADR-014).
- Re-review: 14 CLOSED, 1 ACCEPTED (F-012), 1 NOT CLOSED at 05a06a3 (F-007, fixed afterwards, ADR-018); 5 new findings (2 minor, 3 info).
- Reviewer re-checks at 05a06a3: `run_pytest tests/firmware` "295 passed in 1.93s"; ERC "Found 0 violations"; DRC 0/0/0; status counts 17/18/3/4 = 42 match (MEASUREMENT).

Highlight: defect found while fixing F-005. On an I^2C NACK the Adafruit library's `touched()` returned 0x0FFF, i.e. all 12 pads touched, so a lost sensor would have sent 12 Note Ons. The firmware now reads the status registers itself and checks the transaction result (ADR-017). The re-review confirmed the fix (F-005 CLOSED).

Table 18: Review findings, dispositions and re-review verdicts.

ID	Sev.	Finding (one line)	Disposition	Re-review
F-001	major	A wrong JP1 position was called harmless; it exceeds the MPR121 VDD + 0.3 V absolute maximum and back-feeds the module.	FIX: README §5/§6	CLOSED
F-002	major	The ±4 kV "PASS" compared peak current only; pin limit is an ASSUMPTION; resistor not pulse-rated.	FIX: REQ-023 is an estimate, not a pass	CLOSED
F-010	major	REQ-018 was PARTIAL although budget check A3b FAILs (4.40 V < 4.50 V).	FIX: REQ-018 FAIL	CLOSED
F-003	minor	Pot hysteresis defined two ways; offset share misstated.	FIX: REQ-006 restated	CLOSED
F-004	minor	Two held pads can share a note; releasing one sent Note Off.	FIX: reference counting	CLOSED
F-005	minor	Sensor lost during play not detected; missing second sensor never retried.	FIX: sensor health	CLOSED
F-006	minor	Leonardo could run I^2C at 100 kHz after `begin()`.	FIX: `setClock` after every `begin()`	CLOSED

ID	Sev.	Finding (one line)	Disposition	Re-review
F-007	minor	REQ-033 build logs not in the commit.	FIX: commit logs	NOT CLOSED at 05a06a3; fixed after (ADR-018)
F-008	minor	budgets.md stale; coverage.md "USL 201".	FIX: regenerate, reconcile	CLOSED (residue N-002)
F-009	minor	README named an "aprender" button.	FIX	CLOSED
F-011	minor	Latency model omitted the 15 ms pad lockout.	FIX: REQ-014 scoped to first touch	CLOSED
F-014	minor	Terminal pin diagonal 1.27 mm in a 1.3 mm hole; pot slot drilling unclear.	FIX: 1.4 mm drill, slot instruction	CLOSED
F-012	info	Latency rests on an assumed MPR121 status-update model.	ACCEPTED (plan P-2)	ACCEPTED
F-013	info	ERC ignored single-pin labels globally.	FIX at root (J3)	CLOSED
F-015	info	BOM lacked part numbers; strip count wrong.	FIX	CLOSED
F-016	info	AGND relied on the Pico's internal tie without citation.	FIX: cite S-01/S-02	CLOSED
N-001	minor	Automatic restart of a self-reset sensor blocks the loop 16.8–22.1 ms, beyond REQ-016's 10 ms.	ACCEPTED as specified exception (ADR-018)	new
N-002	minor	REQ-016 evidence stale (worst loop ≈1.72 ms incl. ECR poll).	FIXED (ADR-018)	new
N-003	info	INTERFACES had no J3/J4–J7.	FIXED (frozen-3, IF-PCB-AUX)	new
N-004	info	With 2 sensors configured and one absent the LED blinks fast permanently.	ACCEPTED (matches REQ-009)	new
N-005	info	A second pad on an already sounding note is silent (no retrigger).	ACCEPTED (matches REQ-042)	new

What the review did not cover. First review: INTERFACES.json field by field; visual check of the copper PDFs and mirroring; the sampler beyond its MIDI handler; 8 of 9 firmware targets; MPR121 power-on-reset timing. Re-review: compile runs (logs only); copper PDFs and silkscreen visually; this report's LaTeX; the individual test cases of `test_notes_health.py`. Neither review is a hardware test.

9.1 CR-002 independent re-review

An independent reviewer re-reviewed the CR-002 outputs at `8394b41` on branch `change/CR-002` (task T-071): R1–R12 part and footprint, relayout, gap check, pulse-rating calculation, budgets revision 6, REQUIREMENTS draft-6, INTERFACES frozen-4 and the documentation. The harness refused the reviewer's write of the findings file; the manager recorded the findings from the reviewer's hand-back report without changing their substance [`reports/review/CR-002/findings.md`]. The electronics fixes were made in T-073 (committed in c901ad8) [`management/handoffs/electronics-engineer/T-073.md`]; dispositions are recorded in ADR-020 [`management/DECISIONS.md`].

Summary.

- 13 findings: 0 critical, 0 major, 6 minor, 7 info.
- Dispositions: 8 fixed (C2-006 by systems in T-074, committed in 2ba289a), 1 partly fixed with a residual accepted risk (C2-002), 1 accepted risk (C2-005), 2 accepted (C2-012 with a test added, C2-013), 1 no action (C2-004).
- Reviewer re-checks at 8394b41, all PASS (recorded in the findings file; raw outputs in `reports/review/CR-002/`): ERC 0; DRC 0, unconnected 0, parity 0; `check_gap.py` PASS (copper only); `calc_flashover` and `calc_pulse_rated` outputs byte-identical; interface

check 80/80 (committed output stale, 79/80); `pcb_rules_check` 16/16; budgets byte-identical (23.78–37.72 EUR); independent chart readings at 3.3 kΩ 4086 / 1740 / 1469 V for 0414 / 0207 / 0204; drill count 131 equal to the assembly guide; configuration wires 21.1–32.1 mm (MEASUREMENT, design file).

- Reviewer's requirement cross-check: REQ-023 PARTIAL (agreed); REQ-027 PASS; REQ-028/029 PASS with stale evidence text (C2-006); REQ-032 FAIL; REQ-036 should be PARTIAL until C2-001 and C2-007 are fixed (both fixed in T-073).

Highlights. The closest metal of a flat resistor is its end caps, not the copper (C2-002): the 8 kV gap claims now apply to copper only, and the 4 kV end-cap result is PARTIAL on an assumed cap length (Section 6.5). The pulse energy had been overstated about twice (C2-003): 182 mJ, not 364 mJ, with the verdicts unchanged. A metal screw at the old H3 position could bypass R12 (1.94 mm bypass sum, FAIL); H3 moved to (96, 38) mm (15.36 mm, PASS) (CALCULATION, `check_gap.out`).

Table 19: CR-002 re-review findings and dispositions (ADR-020).

ID	Sev.	Finding (one line)	Disposition
C2-001	minor	Assembly guide asked for a lead bend ≥1.5 mm from the body; at 15.24 mm pitch the lead centre is 1.67 mm from the body end (CALCULATION).	Fixed: reworded (Vishay $M_{\min}$ 15.0 mm ≤ 15.24 mm pitch; hold the lead next to the body, test-fit one part).
C2-002	minor	Closest metal of a flat resistor is cap to cap, not the 13 mm copper gap; `check_gap.py` measures copper only; 8 kV gap claims overreach.	Partly fixed, residual accepted risk: cap-to-cap check added; 4 kV PASS for placement error ≤ ±0.25 mm (worst 3.61 mm), FAIL corner 3.13 mm at ±0.5 mm; 8 kV caps FAIL; leads formed on a gauge, caliper check in the plan; 8 kV statements limited to copper.
C2-003	minor	Pulse energy overstated about 2× ($\int V\,dt$ form); rating energy-limited at 3.3 kΩ.	Fixed: 182 mJ ($\int V^2\,dt$); energy-limited below the 5073 Ω knee, plateau 5054 V; verdicts unchanged.
C2-004	info	Independent chart reading confirms 4086 / 1740 / 1469 V; dimensions, LSR37 data and part number agree.	No action.
C2-005	minor	Every E terminal pad is 1.40 mm from the GND edge track, so a strike most likely arcs to GND; the current through the GND trunk to Pico, module and USB was not analysed.	Accepted risk: intentional spark gap; ground bounce 87–339 V for ≈1 ns, ≤1.53 A and ≤0.92 nC per 220 Ω I^2C resistor (CALCULATION on ASSUMPTIONed inductances); arc to GND added to the ESD test; GP26/GP27 HBM rating not confirmed (open).
C2-006	minor	Stale evidence: committed `interface_check.out` 79/80; REQ-028/029, requirements.csv and INTERFACES frozen-4 notes describe 3a3684b.	Fixed by systems (T-074): `sim/systems/*.out`, REQUIREMENTS, INTERFACES frozen-5 (committed in 2ba289a).
C2-007	minor	Assembly guide offered any resistor lead off-cut for JP1 (0.8 mm holes); R1–R12 leads are 0.8 mm.	Fixed: solder or an off-cut of an R13–R16 lead (0.5 mm) only.
C2-008	info	1.0 mm drill for a 0.8 mm lead sits exactly on the 0.2 mm fit threshold; generator comment wrong.	Fixed: R1–R12 drill 1.1 mm (ring 0.55 mm); comment corrected.
C2-009	info	Courtyard gaps 0.06 mm (R row to J2) and 0.09 mm (R12 to H3); M3 head about 0.5–0.9 mm from R12.	Fixed: H3 moved (95, 48) → (96, 38) mm; bypass 1.94 mm FAIL → 15.36 mm PASS; nylon hardware recommended.
C2-010	info	Terminal labels "10" and "11" touch R10/R11 pad 2 (about 0.1 mm).	Fixed: worst silkscreen-to-pad gap 0.43 mm.
C2-011	info	`check_gap.py` worst-pair name not deterministic on ties.	Fixed: tie-break by reference, then pad.

ID	Sev.	Finding (one line)	Disposition
C2-012	info	I^2C relayout under the module (SDA 3.2 mm from electrode pads over about 31 mm) not analysed.	Accepted, test added: I^2C traffic against baseline noise (V-27).
C2-013	info	IEC 60664-1 F.2 values recalled; the 3.5 mm rule comes from model M4 ($\eta = 0.3$).	Accepted: F.2 kept as a cross-check marked "recalled" (EE-S25).

Re-checks after the fixes (T-073, c901ad8). ERC "ERC messages: 0 Errors 0 Warnings 0"; DRC "Found 0 DRC violations", "Found 0 unconnected pads"; "SUMMARY: 16 checks, 16 PASS, 0 FAIL" (home-etch rules); "SUMMARY: 80 checks, 80 PASS, 0 FAIL" (interfaces) [`reports/electronics/evidence/T-073/`]; `check_gap.out` "SUMMARY: copper PASS (4 kV rule, (a) and (b)); end caps PARTIAL at 4 kV over the swept ASSUMPTIONS; mounting metal PASS". These fixes were reviewed independently in the final re-review T-079, which verified the dispositions C2-001 to C2-013 (Section 9.3).

What the re-review did not cover. The findings file does not list exclusions. It is a design-file and document review: nothing in it is a hardware measurement, and it did not see the T-073 fixes or the systems update T-074; those were covered by the final re-review (Section 9.3).

9.2 Manager PCB design review (analisar-pcb) of v1.2

After the CR-002 re-review fixes, the manager reviewed the v1.2 board with the `analisar-pcb` PCB design review procedure. The review is not independent: the manager also directs the design. The electronics engineer acted on its five items in T-076 (committed in a216c1b); systems updated the checks, budgets, interfaces (frozen-6) and requirements (draft-8) in T-077 (committed in f457bfe). Dispositions are recorded in ADR-021 and its addendum [`management/DECISIONS.md`] [`management/handoffs/electronics-engineer/T-076.md`] [`management/handoffs/systems-engineer/T-077.md`] [`reports/electronics/analysis.md`] (§6).

Table 20: Manager PCB design review of v1.2: items and dispositions (ADR-021).

#	Item	Change and evidence	Disposition
1	JP1 pads 0.60 mm apart on a 2.54 mm pitch: one solder blob can join all three pads (3V3 on both S1 and S7).	Pitch 3.81 mm, pads 1.94×2.4 mm, drill 0.8 mm: pad gap 0.60 → **1.87 mm** on both sides, pads 1 to 3 5.68 mm apart (MEASUREMENT, design file); check "ADR-021 JP1 pad gap >= 1.8 mm" PASS.	Fixed. Bridge with solder or an R13–R16 lead off-cut.
2	J2 screw-terminal pads round 3.0 mm: little copper to resist peeling under screw torque on a home-etched board.	Pads 3.0×4.5 mm oval (pad 1 rectangular), drill 1.4 mm, copper offset 0.75 mm towards the back of the block: copper **7.1 → 11.6 mm²** per pad (CALCULATION). Offset rather than centred so that the Ex to GND edge-track spark gap stays 1.40 mm (a centred oval would give 0.65 mm); adjacent pads 2.08 mm, ring 0.80 mm (MEASUREMENT); `check_gap.out` identical to T-073.	Fixed. Peel strength not measured (V-28).
3	RV1 metal tabs unconnected; wiper unfiltered.	Tabs on AGND; C1 100 nF X7R 50 V, KEMET **C322C104K5R5TA** (5.08 mm leads; the C320 has 2.54 mm leads, SOURCE EE-S29), wiper to AGND. $\tau \leq$ **0.38 ms** worst against the 30 ms read period: settling 2.89 ms, margin **10.4**× (CALCULATION, Section 6.7). Interface check rule changed to AGND membership: 85/85 PASS.	Fixed. Cost +0.20–0.90 EUR (ASSUMPTION) accepted; REQ-032 stays FAIL.

#	Item	Change and evidence	Disposition
4	C2 1 µF from +3V3 to GND at the MPR121 module.	Not added: no single-layer route without running 3V3 next to the H2 fixing metal and a GND track about 48 mm long 1.1 mm from all twelve ELE pins; the ≈100 mm loop would decouple little; both module revisions have their own regulator and capacitors (SOURCE EE-S02/EE-S03).	Rejected, recorded as not needed.
5	Silkscreen overlaps: R16 and SW1 designators overlapped; R14 designator 0.16 mm from R13 pad 1.	Designators moved; JP1 legend line pitch 1.4 → 1.55 mm; `check_silk.py` now also checks footprint designators and text-to-text: worst text-to-pad 0.35 mm, text-to-text 0.11 mm, PASS (MEASUREMENT, design file).	Fixed.

Re-checks after the changes (a216c1b, f457bfe). ERC "ERC messages: 0 Errors 0 Warnings 0"; DRC "Found 0 DRC violations", "Found 0 unconnected pads", "Found 0 Footprint errors" [`reports/electronics/evidence/T-076/`]; "SUMMARY: 19 checks, 19 PASS, 0 FAIL" (home-etch rules); "SUMMARY: 85 checks, 85 PASS, 0 FAIL" (interfaces); budgets "SUMMARY: 30 checks, 25 PASS, 5 FAIL" (the five FAILs are the pre-existing A3b VBUS row, three 100 kHz latency rows and the BOM cost) [`sim/systems/`]. A mutation test on a scratch copy of the board (an SW1 GND pad moved to AGND, JP1 pad 2 shifted) made the new AGND and JP1 checks FAIL as intended (T-077). These changes were reviewed independently in the final re-review T-079 (Section 9.3); none of them has been checked on hardware.

9.3 CR-002 final independent re-review (T-079)

An independent reviewer reviewed the whole CR-002 state at `be9d1f2` on branch `change/CR-002` (task T-079), including the T-073/T-074 fixes and the manager PCB review follow-up of ADR-021 (T-076/T-077) and this report at v1.2 (T-078). As in T-071, the harness refused the reviewer's write; the manager recorded the hand-back content without changing its substance [`reports/review/CR-002-final/findings.md`]; raw outputs are in `reports/review/CR-002-final/`. Dispositions are recorded in ADR-022 [`management/DECISIONS.md`]; the fixes were made in T-080 (electronics) and T-081 (systems), committed in a51dfff [`management/handoffs/electronics-engineer-T-080.md`] [`management/handoffs/systems-engineer-T-081.md`].

Summary.

- Verdict: **release-ready yes** (design-file release v1.2); 7 findings: 0 critical, 0 major, 4 minor (F2-001 to F2-004), 3 info (F2-005 to F2-007).
- Dispositions: 6 fixed, 1 accepted risk with a test and a part specification added (F2-002, risk R-011).
- Reviewer re-checks at be9d1f2 (recorded in the findings file): ERC 0; DRC 0, unconnected 0, parity 0; generators reproduce the committed design (schematic 0 diff lines, PCB 179/179 blocks, identical geometry); B.Cu gerber and drill match the PCB (133 holes); interface check 85/85; `pcb_rules_check` 19/19; budgets 30 checks, 25 PASS, 5 FAIL; 11 electronics calculation scripts byte-identical; C1 RC recomputed: τ 0.380 ms, settling 2.89 ms against `POT_READ_MS` 30 ms, margin 10.4× (CALCULATION); report PDF 39 pages. Dispositions C2-001 to C2-013 verified (the C2-002 residual wording became F2-004).
- Reviewer's requirement cross-check: REQ-006 PARTIAL; REQ-023 PARTIAL; REQ-027/028/029 PASS; REQ-032 FAIL; REQ-036 PASS (all as recorded).
- Not covered: hardware, ESD, peel strength and knob fit; the RK09K and current KEMET TA datasheets were not opened.

Table 21: CR-002 final re-review findings and dispositions (ADR-022).

ID	Sev.	Finding (one line)	Disposition
F2-001	minor	C1 lead 0.48–0.61 mm (SOURCE EE-S29); the 0.8 mm drill leaves 0.19 mm at the largest lead, below the 0.2 mm fit rule; `calc_mechanical` used the nominal lead.	Fixed (T-080): drill 0.9 mm, fit 0.29 mm, ring 0.55 mm (CALCULATION); calculation uses the largest lead.
F2-002	minor	RV1 frame (and a metal shaft) on AGND gives a knob/shaft discharge a direct path into Pico pin 33; no analysis, risk or test; shaft material not fixed.	Accepted risk R-011 (OPEN): plastic knob $\leq$ 15 mm on an insulated shaft specified; knob/shaft strike added to the REQ-023 test (step 2b, V-29).
F2-003	minor	REQ-032 said the C1 cost was not yet accepted, while the ADR-021 addendum accepts it.	Fixed (T-081): REQ-032 evidence cites ADR-019 D-3 and the ADR-021 addendum; status FAIL.
F2-004	minor	Assembly guide said end caps "at least 4 mm apart"; at $\pm$0.25 mm the worst is 3.61 mm (3.13 mm at $\pm$0.5 mm).	Fixed (T-080): $\approx$4.0 mm centred, $\geq$3.6 mm at $\pm$0.25 mm, $\approx$3.1 mm at $\pm$0.5 mm; lead-forming gauge required.
F2-005	info	C1 up to 6.6 mm tall, 7.8 mm from the shaft: knobs wider than about 15.5 mm overhang C1; substitute body size not limited.	Fixed (T-080): knob $\leq$ 15 mm; substitute leads $\leq$ 0.7 mm, body $\leq$ 5.08 $\times$ 3.18 mm, height $\leq$ 6.6 mm.
F2-006	info	No pass/fail check enforced the 1.40 mm Ex–GND spark gap (a flipped J2 offset gives 0.65 mm and still passes DRC).	Fixed (T-081): `pcb_rules_check` spark-gap check, 12/12 pads at 1.400 mm; mutation tests FAIL at 0.650 and $-$0.100 mm.
F2-007	info	`analysis.md` §6 typo; C1 3D model at 5.00 mm pitch.	Fixed (T-080): typo corrected; model scaled to 5.08 mm (render only).

Re-checks after the fixes (T-080/T-081, a51dfff). ERC "ERC messages: 0 Errors 0 Warnings 0"; DRC "Found 0 DRC violations", "Found 0 unconnected pads", "Found 0 Footprint errors" [`reports/electronics/evidence/T-080/`]; "SUMMARY: 20 checks, 20 PASS, 0 FAIL" (home-etch rules, now with the spark-gap check); "SUMMARY: 85 checks, 85 PASS, 0 FAIL" (interfaces); budgets "SUMMARY: 30 checks, 25 PASS, 5 FAIL" (the same five known FAILs) [`sim/systems/`]. Geometry diff against e9309d8 (same design as be9d1f2): only the C1 drill (0.800 $\rightarrow$ 0.900 mm) and the C1 3D model scale changed [`reports/electronics/evidence/T-080/geo_diff.txt`]. By ADR-022 the T-080/T-081 changes (a drill size, wording and a new check) were not re-reviewed independently again; the manager reproduced the checks above. None of this is a hardware test.

10 Limitations

v1.3 limitations. In addition to the items below: the pulse capability of the v1.3 electrode resistors is NOT PROVEN (ADR-026); hand-soldering the Pico castellations and the butted module pins has not been tried; the module height, pin length and cutting tolerance are ASSUMPTIONs (first-build list, Section 3.6); Pico W and Pico 2 W are not supported on the PCB; the D3 screw-force limit of 100 N is unsourced; shipping cost is NOT CONFIRMED.

No hardware tests. Nothing has been built, flashed, powered or measured. Building and measuring hardware is out of scope for this project (risk R-004, "Certain", "High") [`management/RISKS.md`]. Every [PHYSICAL] requirement is therefore PARTIAL or NOT RUN, and no statement in this report means that the hardware works.

Analysis-only results. The following results rest only on calculation, simulation, design-rule checks or host tests: latency and jitter (REQ-014/015); VBUS range and current (REQ-018/019); I^2C rise time and logic levels (REQ-026/041, REQ-024); electrode ESD (REQ-023); sensitivity with the series resistors (SIMULATION); ERC/DRC/parity and home-etch rules;

firmware behaviour (host tests on a copy of the loop order); sampler behaviour (headless Chrome with a stubbed Web MIDI API).

Open FAILs and accepted risks. REQ-018 (4.40 V low-power-port VBUS minimum below the Pico's recommended 4.50 V; documented deviation, ADR-016); REQ-032 (BOM upper bound 38.62 EUR > 30 EUR after CR-002 and ADR-021, prices mostly ASSUMPTION; increases accepted for ESD safety, ADR-019 D-3, and for the C1 wiper filter, ADR-021 addendum); REQ-040 (wrong JP1 position exceeds MPR121 input ratings on an unpowered module; accepted risk R-007); R-006 (at 8 kV contact the 7.27 kV across each resistor exceeds its 4.08 kV rating, and the pin current exceeds the inferred limit by 1.65×; accepted, ADR-019 D-1). CR-002 re-review (ADR-020): end-cap gap at 4 kV PARTIAL, resting on an assumed cap length (C2-002); Ex ↔ GND spark gap with an estimated 87–339 V ground bounce for ≈1 ns and no confirmed HBM rating for GP26/GP27 (C2-005). Final re-review (ADR-022): knob/shaft discharge path into AGND not analysed, plastic knob ≤ 15 mm on an insulated shaft specified and a test step added (R-011, F2-002).

Main assumptions (each to be retired by the plan in Section 11):

- Component prices (BOM, REQ-032) [`reports/systems/budgets.md`].
- RV1: a metal shaft is in contact with the frame and tabs (RK09K datasheet not opened); an insulated-shaft RK09K variant exists (not confirmed); 0.27 mm plan clearance of a 15 mm knob over C1 is enough [`management/handoffs/electronics-engineer-T-080.md`] [`management/RISKS.md`] (R-011).
- Resistor end-cap length 1.5–3.5 mm, body length ≥ 11.0 mm, placement error ≤ ±0.5 mm; track inductance 0.7–1.0 nH/mm and socket plus pin 3–10 nH [`management/handoffs/electronics-engineer/T-073.md`].
- Electrode capacitance: objects up to about 500 pF (REQ-037); 1 m lead 30–50 pF; hand 100–300 pF [`reports/electronics/analysis.md`].
- ESD: ±4 kV target following IEC 61000-4-2 level 2 (the standard was not read); MPR121 pin limit 1.33 A inferred from HBM 2 kV; electrode discharge model; the breakdown models of the gap rule and the IEC 60664-1 Table F.2 values (recalled, not re-read, EE-S25); the resistor chart reading (±5 %); a 1.2/50 µs rating taken as conservative for the ≈0.54 µs pulse; the 3k3 MBE 0414 price equal to its sibling values (EE-S26).
- Board currents (upper bounds), Schottky diode drop and regulator efficiency [`reports/systems/budgets.md`].
- I^2C bus capacitance; Pico pin capacitance; in the electronics calculation, RP2040 VIL 0.8 V and VOL 0.4 V and BSS138 $R_{DS(on)} \leq 10\,\Omega$ (T-055).
- MIC5225 typical dropout from a search extract only [`reports/electronics/analysis.md`].
- Firmware timing: loop period "well under 15 ms", button bounce < 10 ms, header and pot pin sizes for the 1.0 mm holes [`reports/firmware/coverage.md`].

Process limitations. Requirements are draft-14 (T-106), not baselined. The independent re-review of the CR-002 changes is reported in Section 9.1; the later manager PCB review and its fixes (ADR-021, Section 9.2) were reviewed independently in the final re-review T-079 (verdict release-ready, findings F2-001 to F2-007, all dispositioned in ADR-022; Section 9.3). The T-080/T-081 fixes of those findings were reproduced by the manager, not re-reviewed independently (ADR-022). The committed systems outputs are current at a51dfff (interface check 85/85, homeetch rules 20/20, budgets 25 PASS and 5 FAIL of 30). Some user-documentation statements (prices, OS behaviour, etching process values) are not traced to a source (documentation handoffs T-030 to T-035).

10.1 Risk register

From [management/RISKS.md]. CR-002 rewrote R-006 and added R-008 to R-010; ADR-021 added no risk; the final re-review added R-011 (ADR-022).

ID	Risk	L / I	Mitigation	Status
R-001	Two Adafruit 1982 revisions with different power-row pin order; wrong wiring shorts 3V3 to GND	M / H	Configuration wires and inspection table	OPEN
R-002	Large conductive objects may exceed the MPR121 range or be unstable	M / M	AUTOCONFIG; threshold sweep; validation plan	OPEN
R-003	Home-etch defects (bridges between 2.54 mm pads)	M / M	0.6 mm clearance rule; continuity checklist	OPEN
R-004	Nothing tested on real hardware (out of scope)	Certain / H	Physical validation plan; no readiness claims	OPEN
R-005	Limited disk space on the host	M / M	Install only what is needed	OPEN
R-006	ESD at 8 kV: 7.27 kV across each 3k3 (CALCULATION) exceeds the 4.08 kV rating of the MBE 0414 (SOURCE EE-S17b); pin current 2.20 A vs inferred 1.33 A (1.65×). Copper gaps meet the 8 kV 7.5 mm rule; the end caps do not (T-073, finding C2-002)	L / M	Scope stays ±4 kV (ADR-008, ADR-019 D-1); later option Vishay LSR37 (same footprint, 12 kV), deferred for tolerance and drift; IEC test in the plan	ACCEPTED
R-007	Wrong JP1 position puts +3V3 on ADDR (QT) or IRQ (original) of an unpowered module	L / L–M	Silkscreen, manual warning, LED error pattern (ADR-009)	ACCEPTED
R-008	v1.2 relayout lengthens the two J1 → S configuration wires to about 25–35 mm: a loose wire can be pushed onto the wrong S hole or strain its joint. About 1.8 pF per line (CALCULATION, 50 pF/m ASSUMPTION) vs the 115.2 pF limit; a wrong hole is limited to 15.5 mA by R15/R16	L / L	Wiring table, pre-power inspection; documentation says to route the wires flat and short; physical check in the plan	OPEN
R-009	End-cap flashover around R1–R12 at 4 kV depends on lead forming and on the unsourced end-cap length (C2-002): cap-to-cap worst 3.61 mm at ±0.25 mm placement error (PASS against 3.5 mm), 3.13 mm at ±0.5 mm (FAIL) (CALCULATION on ASSUMPTIONs)	M / M	Leads formed on a gauge; caliper check of one MBE 0414 before the ESD test (V-26); REQ-023 stays PARTIAL (ADR-020)	ACCEPTED
R-010	Electrode pads 1.40 mm from the GND edge track (intentional spark gap, C2-005): estimated 87–339 V ground bounce for ≈1 ns, ≤1.53 A and ≤0.92 nC per I^2C line (CALCULATION on ASSUMPTIONs); no HBM rating found for GP26/GP27	L / M	Spark gap kept (ADR-020; also kept by the J2 pad change of ADR-021) and enforced by `pcb_rules_check` (1.40 ± 0.05 mm, ADR-022); firmware recovers a lost sensor; ESD test watches for arcs, I^2C errors and resets (V-25)	OPEN
R-011	Knob/shaft ESD path into AGND (F2-002): the RV1 tabs and frame are on AGND (ADR-021) and a metal shaft is assumed to touch the frame (ASSUMPTION); a discharge to a metal shaft or conductive knob enters AGND at Pico pin 33 with no series resistor. Not analysed numerically (no peak current, flashover path or RP2040 ADC/AGND ESD rating)	L / M	Plastic knob ≤ 15 mm covering the whole shaft, insulated-shaft RK09K variant where available; knob/shaft strike in the REQ-023 test (step 2b, V-29); fallbacks (tabs unconnected or on GND) only if that test fails	OPEN

11 Physical validation plan

v1.3 note. The rows below were written for v1.2. Rows that name v1.2 parts (for example V-26 on the MBE0414 resistor) do not apply to v1.3; the v1.3 first-build measurements are in Table 6, and the REQ-023 ESD series (rows on REQ-023 below) applies to v1.3 with the RC2512 values and the plan-B trigger of Section 3.4.

No part of the hardware has been measured. Each row is a physical test still to run. Pass criteria are the requirement thresholds in [`management/REQUIREMENTS.md`]; where a requirement gives no threshold, the analysis prediction is the criterion and is labelled. Items P-1 to P-6 are the systems plan [`management/handoffs/systems-engineer/T-011.md`]; hardware items from firmware are in [`reports/firmware/coverage.md`]; electronics next tests are in the T-050 to T-055, T-073, T-076 and T-080 handoffs; V-29 is step (2b) of the REQ-023 plan added after the final re-review (T-081, ADR-022). All rows are NOT RUN.

Table 23: Physical validation plan (all NOT RUN).

ID	What to measure	How / instrument	Pass criterion	REQ
V-01 (P-1)	Object plus lead capacitance	LCR meter on typical objects (bare pad, fruit, glass of water) with 20–50 cm leads	Touch detected after recalibration from about 20 pF to about 500 pF (ASSUMPTION range in REQ-037)	REQ-037
V-02 (P-2)	Touch-to-USB latency and jitter	Logic analyser on a test electrode, SDA/SCL and USB	Worst case $\leq$ 15 ms; jitter $\leq$ 10 ms, for $C \leq 100$ pF	REQ-014, 015
V-03 (P-3)	VBUS current per Pico variant	USB inline current meter; Pico, Pico W, Pico 2, Pico 2 W	$\leq$ 100 mA	REQ-019
V-04	VBUS range	Adjustable bench supply at 4.40 V and 5.50 V at the connector	Device enumerates and plays at both ends	REQ-018
V-05 (P-4)	I^2C rise time and low levels, PCB and breadboard	Oscilloscope on SDA/SCL at the Pico and at the module	Rise $\leq$ 300 ns at 400 kHz; low levels below VIL (0.93 V at the MPR121, 0.8 V at the Pico)	REQ-026, 041
V-06	Misplaced data wire current	Wire J1 SDA/SCL to a supply hole; current meter in series	$\leq$ 15.5 mA per line (CALCULATION); within RP2040 IO limits	REQ-024
V-07 (P-5)	Host enumeration	Windows 10, Windows 11, macOS 12 or later; MIDI monitor	No vendor driver; name "FrutaMIDI"; Note On/Off and poly aftertouch on channel 1 only	REQ-011, 012, 013
V-08 (P-6)	False triggers and crosstalk	10 min idle with all 12 electrodes on objects; then touch each pad and watch neighbours	No false Note On; no crosstalk	REQ-038
V-09	Electrode readings with and without R	`SERIAL_DEBUG` deltas per object, with the 3.3 kΩ resistor and with it bridged	Touch detected on every object; delta ratio $\geq$ 0.904 (SIMULATION prediction)	REQ-037
V-10	ESD robustness	IEC 61000-4-2 contact discharge ±4 kV on each terminal, at a test lab	No permanent damage: MPR121 still responds, all pads still detect touch, baseline drift $<$ 1 count (T-060)	REQ-023
V-11	Series resistor after ESD	Watch for flashover across R1–R12 during V-10; measure R1–R12 afterwards	No flashover; each resistance within 1 % of its pre-test value (T-060)	REQ-023
V-12	JP1 bridge and etch gaps	Visual inspection with a magnifier and continuity after etching and bridging	Bridge only on the chosen side; the 1.87 mm gap on the other side clear (ADR-021) and 0.6 mm track gaps clear; no short to GND in any position	REQ-025, 039

ID	What to measure	How / instrument	Pass criterion	REQ
V-13	Real loop period	Toggle a spare GPIO each loop; oscilloscope	No blocking > 10 ms during play, except user recalibration and the lost-sensor restart (ADR-018)	REQ-016
V-14	Recalibration duration	GPIO toggle around `recalibrate()`	Recorded per sensor; compare with 16.8–22.1 ms (CALCULATION); no stuck notes afterwards	REQ-008, 002
V-15	Pot ADC noise on the PCB	Log `analogRead` at each zone border for 10 s	Peak-to-peak noise < 41 counts	REQ-006
V-16	Blink with live pads	Touch pads while the LED blinks scale 6; press again during a blink	Notes sent during the blink; pattern restarts with the new scale	REQ-007, 016
V-17	Button bounce and long press	Oscilloscope on the button pin; press tests	Bounce < 10 ms (ASSUMPTION); long press acts after > 800 ms	REQ-007, 008
V-18	Sensor absent or lost, and LED	Power without the module; then with it; unplug the module during play; every Pico board	No MIDI from the missing sensor, its notes released, LED at 5 Hz, retry every 2 s; 2 slow blinks when found; LED lit while touched (W boards via CYW43)	REQ-009, 010
V-19	MPR121 configuration and burst reads	Read back 0x7B–0x7F after start-up; logic analyser during play	Values as configured; 2 transactions per sensor per aftertouch pass; burst data equal to per-electrode reads	REQ-004, 037
V-20	Sampler in real browsers	Chrome and Edge on Windows and macOS with the device	Pads 36–47 sound; velocity 0 acts as Note Off; learn swaps notes	REQ-034
V-21	Sampler with a busy device	Open the device in a DAW on Windows first, then the sampler	Sampler reports the device as busy	REQ-034
V-22	Shared notes	Set an octave so two pads share a note; touch both, release one, then the other	One Note On; Note Off only after the last release; aftertouch = the stronger pad	REQ-042
V-23	v1.2 configuration wires (R-008)	Inspect the two J1 $\rightarrow$ S wires before power; continuity J1 SDA/SCL to the S hole of the module revision	Each wire on the correct S hole (S5/S6 original, S3/S4 STEMMA QT), flat and mechanically secure	REQ-024, 025
V-24	R1–R12 fitted as designed	Inspect the resistor row after soldering; caliper on one resistor	Lying flat, part MBE04140C3301FC100, leads 15.24 mm apart, no lead bent towards a neighbour; leads formed on a gauge, the two straight lengths equal within about 0.25 mm (T-073)	REQ-023
V-25	Arc path and I^2C upsets during ESD (C2-005)	During V-10, watch each terminal for an arc to the GND edge track; after every strike check for I^2C NACKs (logic analyser on SDA/SCL), the lost-sensor LED blink, Pico resets (USB re-enumeration) and stuck notes (MIDI monitor)	Arc, if any, to the GND track only (not across Rx); no reset, no lost sensor and no stuck note after any strike; any I^2C error recorded and reported (ADR-020)	REQ-023
V-26	MBE 0414 body and end-cap length (C2-002)	Caliper on one resistor before fitting: body length L and cap length L_c; then re-run `check_gap.py` section (d) with the measured values	$L_c \leq 3.0$ mm (4 kV rule met at every swept L and placement error, CALCULATION); otherwise the recomputed cap gap ≥ 3.5 mm at the measured placement error	REQ-023

ID	What to measure	How / instrument	Pass criterion	REQ
V-27	I^2C traffic against electrode baseline noise (C2-012)	`SERIAL_DEBUG` deltas per electrode with the bus idle between reads versus continuous I^2C polling, electrodes on typical objects	No false Note On; noise with traffic increase of at most 3 counts in the per-electrode delta (ASSUMPTION threshold, T-074)	REQ-038
V-28	ADR-021 parts and pads	Test-fit C1 (5.08 mm leads, 0.48–0.61 mm) in its 0.9 mm holes; fit the 15 mm plastic knob and check it clears C1 (0.27 mm plan clearance, CALCULATION); bridge JP1 with an R13–R16 lead off-cut over the 1.87 mm gap; tighten and re-tighten every J2 screw; with the pot fitted, log `analogRead` during slow and fast turns	C1 seated flat without force; knob turns over its full travel without touching C1; JP1 bridge on one side only, no continuity to the other side; no J2 pad lifted or cracked; the octave follows the knob with no visible lag (settling $\leq$ 2.89 ms, CALCULATION)	REQ-006, 025, 029
V-29	Knob/shaft ESD strike (REQ-023 step 2b, R-011, F2-002)	During V-10, with the specified plastic knob ($\leq$ 15 mm) on the insulated shaft: $\pm$4 kV contact discharge on the knob top and side, then with the knob removed on the shaft tip; record I^2C errors, the sensor-lost blink, USB disconnect or reset, and the pot CC value before and after; record the knob and shaft actually fitted	No permanent damage: every pad still detects touch and sends MIDI, and RV1 still sweeps the full CC range; temporary upsets that recover without a power cycle are recorded, not a fail; a non-insulating knob or shaft makes the result not representative. If it fails: R-011 fallbacks (tabs unconnected or on GND)	REQ-023

12 Reproducibility

12.1 Toolchain

From [`management/TOOLCHAIN.md`] (MEASUREMENT by command, 2026-10-30) and the task handoffs:

- Host: macOS 26.5.2, arm64, Apple M4.
- KiCad 10.0.4 (kicad-cli and pcbnew Python 3.9.13).
- arduino-cli 1.5.1; arduino-pico core rp2040 6.3.1; AVR core 1.8.8; Adafruit MPR121 1.2.1, BusIO 1.17.4, MIDI Library 5.0.2, MIDIUSB 1.0.5.
- Python 3.9.6 with pytest 8.4.2 (project `.venv`); node v26.4.0; playwright-core 1.64.1 and Google Chrome 156.0.8078.74 (T-020).
- TinyTeX-1 v2026.10, pdfTeX 1.40.29 (TeX Live 2026), with siunitx, enumitem and caption added (documentation handoff T-006-DOC/T-031).
- MCP adapter `eng-tools` (`scripts/mcp/server.mjs`, @modelcontextprotocol/sdk 1.32.2).

12.2 How to rebuild

- Hardware: `sh hardware/scripts/build_all.sh` regenerates the schematic and PCB, runs ERC and DRC with schematic parity (fails on violations) and exports `hardware/fabrication/`.
- Firmware build matrix: eng-tools `firmware_compile`, or `arduino-cli compile` with the FQBN `rp2040:rp2040:<board>:usbstack=tinyusb` for rpipico, rpipicow, rpipico2,

rpipico2w, each without and with `--build-property compiler.cpp.extra_flags=-DFRUTAMIDI_PCB`, plus `arduino:avr:leonardo` [`management/INTERFACES.json`].

- Host tests: eng-tools `run_pytest` with `tests/firmware` and `tests/host`.
- Systems checks: `sim/systems/budgets.py`, `interface_check.py`, `pcb_rules_check.py`; electronics calculations `sh sim/electronics/calc/run_all.sh`.
- Report: `sh reports/report/build.sh` (regenerates the requirements table from the CSV, then three pdflatex passes), or eng-tools `latex_build` with `reports/report/frutamidi_report.tex`.

Released file checksums: `releases/V1_BASELINE/SHA256SUMS` for the previous release (verify with `shasum -a 256 -c SHA256SUMS`). Release V1.2 (tag `v1.2.0`) has not been made; its checksum file will be created at release. A document cannot contain its own checksum, so this report's hash is listed there. The board figures in `reports/figures/` are byte-identical copies of `hardware/fabrication/images/` at a51dfff (sha256 in the T-082 handoff).

13 Change history

Baseline: release V1_BASELINE, git tag `v1.1.0-baseline` (048023c, 2026-11-04), board label "FrutaMIDI v1.1", interfaces frozen-3. V1.2 (CR-002) was released as git tag `v1.2.0`. Current: FrutaMIDI v1.3 on branch `change/CR-003` at f6693c9 (review fixes T-103/T-104, interfaces frozen-8 with a frozen-9 proposal, requirements draft-14), not released. Sources: `git log v1.1.0-baseline..a51dfff` and `git log 8e963d0..f6693c9`, the commit messages quoted below, [`changes/CR-002.md`], [`changes/CR-003.md`], ADR-019 to ADR-028 [`management/DECISIONS.md`].

Commit	Change	Effect on nets, copper and checks
21a4919	Schematic readability: A3 sheet, framed functional blocks, electrode chain drawn with wires. Layout of the schematic only.	Netlist identical (127 nets, compared node by node); PCB unchanged; ERC 0, DRC 0, parity 0 (commit message).
b1e4bf5	The unconnected BOM-only socket symbols J4–J7 are removed; the female-header sockets become a Socket field of A1/A2; the BOM gains a Socket column.	Netlist unchanged apart from the removed J4–J7 no-connect nets; PCB copper identical (tracks and pads compared); ERC 0, DRC 0, parity 0 (commit message).
130e754	CR-002 opened: electrode resistor flashover and pulse rating; supersedes candidate CR-001.	Documents only.
3a3684b	CR-002 T-060: R1–R12 become Vishay MBE04140C3301FC100 (DIN0414, lying flat, 15.24 mm pitch); relayout (module A2 15.1 mm up and 2.54 mm right; J1, JP1 upright, R13–R15 right of the module; H4 moved); analysis.md §5; ADR-019.	ELE–E gap 0.94 → 13.04 mm, E–other net 1.99 → 10.92 mm (MEASUREMENT, design file); nets and pinout unchanged; ERC 0, DRC 0, parity 0; pcb_rules_check 16/16.
8394b41	CR-002 T-070: REQUIREMENTS draft-6, INTERFACES frozen-4, budgets revision 6 (core BOM 23.78–37.72 EUR, CALCULATION), risks R-006 rewritten and R-008 added; board label "FrutaMIDI v1.2".	Label text only on the board; gap check unchanged (re-run in T-072).
dcdb1fc	CR-002 T-072: this report, the user manual and figures updated for v1.2.	Documents only.

Commit	Change	Effect on nets, copper and checks
1247c44	CR-002 T-071: independent re-review findings C2-001 to C2-013 recorded (Section 9.1).	Documents only.
c901ad8	CR-002 T-073: review fixes. End-cap (cap-to-cap) and mounting-metal checks added to `check_gap.py`; pulse energy corrected to 182 mJ; R1–R12 drill 1.0 → 1.1 mm; H3 moved (95, 48) → (96, 38) mm; terminal labels moved off the pads; Ex–GND gap kept as a spark gap with a ground-bounce estimate (`calc_gnd_bounce.py`); assembly guide reworded (lead forming on a gauge, JP1 off-cut, nylon screw at H3, flat configuration wires); ADR-020.	Nets and pinout unchanged; copper gaps 13.04 / 10.92 mm unchanged; end caps 4 kV PARTIAL (worst 3.61 mm at ±0.25 mm); ERC 0, DRC 0, unconnected 0; pcb_rules_check 16/16; interface check 80/80 (MEASUREMENT / CALCULATION, T-073 handoff).
2ba289a	CR-002 T-074/T-075: REQUIREMENTS draft-7, INTERFACES frozen-5, risks R-009 and R-010; this report (CR-002 re-review section, ESD section, validation rows V-25 to V-27), README, user manual and figures updated after T-073.	Documents and check outputs only.
565ffb5	Report build fix: requirements text without U+222B; report rebuilt.	Documents only.
a216c1b	T-076, manager PCB review follow-up (ADR-021): JP1 pitch 2.54 → 3.81 mm; J2 pads 3.0 mm round → 3.0 × 4.5 mm oval, offset 0.75 mm towards the block back; RV1 mounting tabs on AGND; new C1 100 nF KEMET C322C104K5R5TA from the pot wiper to AGND; silkscreen overlaps removed; `calc_pot_rc.py` added; assembly guide updated.	New part C1 and new AGND members (RV1 tabs, C1 pin 2); other nets and pinout unchanged. JP1 gap 0.60 → 1.87 mm; J2 copper 7.1 → 11.6 mm^2 per pad; Ex–GND 1.40 mm unchanged; 0.8 mm holes 11 → 13 (C1 at 0.8 mm, changed to 0.9 mm in a51dfff), 133 holes in total; ERC 0, DRC 0, parity 0 (MEASUREMENT / CALCULATION, T-076 handoff).
f457bfe	T-077: interface check AGND membership rules (replacing `pcb.A1.AGND_isolated`), home-etch checks for the JP1 gap and J2 pads, budgets revision 7 (check A7, C1 row), INTERFACES frozen-6, REQUIREMENTS draft-8.	Checks only: interface check 85/85, pcb_rules_check 19/19, budgets 30 checks, 25 PASS, 5 FAIL; core BOM 23.98–38.62 EUR (CALCULATION).
be9d1f2	T-078: this report (manager PCB review section, C1 and AGND, cost, interfaces), README, user manual and figures updated after T-077.	Documents only.
e9309d8	T-079: final independent re-review findings F2-001 to F2-007 recorded, verdict release-ready (Section 9.3).	Documents and reviewer raw outputs only.

Commit	Change	Effect on nets, copper and checks
a51dfff	T-080/T-081, final re-review fixes (ADR-022): C1 drill 0.8 → 0.9 mm (fit 0.29 mm, ring 0.55 mm) and C1 3D model at 5.08 mm; `calc_mechanical` judges C1 on its largest lead; assembly guide: end-cap wording (≈4.0 / ≥3.6 / ≈3.1 mm, lead-forming gauge), plastic knob ≤ 15 mm on an insulated shaft, C1 substitute limits; spark-gap check in `pcb_rules_check`; risk R-011; REQUIREMENTS draft-9 (REQ-023 step 2b, REQ-032 evidence); INTERFACES frozen-7.	Nets and pinout unchanged; only C1 drill and model scale changed in the board (geometry diff); drill file 0.8 mm 11, 0.9 mm 11, 133 holes in total; ERC 0, DRC 0, parity 0; pcb_rules_check 20/20; interface check 85/85; budgets 30 checks, 25 PASS, 5 FAIL (MEASUREMENT / CALCULATION, T-080/T-081 handoffs).
(T-082)	This report (final re-review section, C1 holes, knob, R-011, V-29, interfaces frozen-7, requirements table), README, user manual and board figures updated after T-080/T-081.	Documents only.
CR-003: SMD board for home chemical etching (v1.3)		
8e5f595	CR-003 opened: SMD conversion, impact analysis.	Documents only.
8c89555	T-090: SMD feasibility study; ADR-023 user decisions.	Documents only.
f3a718a	T-091: SMD schematic and single-layer layout on F.Cu with 4 holes; SMD checks; assembly guide.	Full re-layout; holes 133 → 4 (MEASUREMENT, CR-003); pin map unchanged.
76027e9	T-092: mounting holes moved for USB support; calculations re-run for the SMD board.	Hole positions only.
410933d	T-093: requirements draft-10, SMD rules and interface checks, budgets, risks R-012 to R-017, INTERFACES frozen-8 proposal; project definition amended.	Checks and documents.
463f18c	T-094: SDA re-routed 3.30 mm from the electrode pads (R-017).	One track moved.
0409dcb, 257fcfa, 68866fb, 04edcc9	T-095 quote comparison; ADR-024 (Mouser BOM), ADR-025, ADR-026 (R1–R12 YAGEO RC2512FK-073K3L, pulse capability not proven, accepted).	Decisions only.
1f858f0, 5c92ca3	T-096/T-097: R1–R12 YAGEO RC2512FK-073K3L, Mouser BOM 13.84 EUR, RV1 PTV09A; TVS clamp study (plan B, not adopted); T-098 footprint rename.	R1–R12 land unchanged.
9505245, d7da0c2	T-099/T-100: systems checks, budgets revision 9, requirements draft-11 and draft-12; ADR-027; INTERFACES frozen-8 frozen.	Checks and documents.
7c3cf6e	T-101: manual and README for v1.3; metrics for v1.3.	Documents only.
fd7140b	T-102: independent review findings F3-001 to F3-015, verdict not ready for the first build (Section 3.5).	Documents only.
f6693c9	T-103/T-104: review fixes (A2 lands outside the module, Pico moved and Pico 2 keep-out, 1206 lands, J1 out of the BOM), systems re-checks, requirements draft-13, ADR-028.	ERC 0, DRC 0, parity 0; check_smd 65/65; pcb_rules_check 26/26 (T-103/T-104 handoffs).
(T-105)	This report (Section 3, v1.3 figures), user manual and root README updated after T-103/T-104.	Documents only; interface_check 121/121.
(T-106, T-107)	Requirements draft-14 (REQ-036 PASS); this report updated to draft-14.	Documents only.

What did not change in v1.2. Firmware, pins and build flags; the J2 terminal row and pinout; the I^2C, USB MIDI and terminal interfaces apart from the R1–R12 part, C1, the AGND members and the JP1 and J2 footprints (ADR-021); the resistor value (3.3 kΩ 1 %)

and therefore the electrode sensitivity analysis; the home-etch design rules (ADR-019 D-4, [management/INTERFACES.json]).

Requirement status after CR-002. Unchanged counts in REQUIREMENTS draft-9 at a51dfff (PASS 17, PARTIAL 18, FAIL 3, NOT RUN 4). REQ-023 stays PARTIAL (no ESD-gun test); REQ-006 stays PARTIAL (C1 settling by calculation only); REQ-032 stays FAIL with a larger overrun (8.62 EUR instead of 0.40 EUR, CALCULATION) [management/REQUIREMENTS.md].

Requirement status after CR-003. Counts in REQUIREMENTS draft-14 (T-106): PASS 18, PARTIAL 18, FAIL 2, NOT RUN 4. REQ-032 moved from FAIL to PASS on the ADR-027 D-1 basis; REQ-036 moved from PASS to FAIL while the documents followed the board (draft-10) and back to PASS in draft-14 after T-105 (document consistency only); REQ-023 stays PARTIAL [management/REQUIREMENTS.md].

A Sources

Source registers: [management/SOURCES.md] (S-0xx), [requirements/SOURCES_SYSTEMS.md] (S-xx) and [reports/electronics/SOURCES_EE.md] (EE-Sxx, marked "Read" or "Search"). The registers hold the full entries, access notes and checksums. Sources cited in this report:

ID	Source
S-01	Raspberry Pi Pico datasheet RP-008307-DS-2 (local copy, sha256 in the register).
S-02	Raspberry Pi documentation, "Pico-series microcontrollers", pin functions. https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html
S-04, S-006, EE-S01	NXP/Freescale MPR121 datasheet, Rev. 4, 02/2013. https://www.nxp.com/docs/en/data-sheet/MPR121.pdf
S-06	Adafruit Eagle board files for the MPR121 breakout (STEMMA QT revision). https://github.com/adafruit/Adafruit-MPR121-PCB
S-11	Adafruit TinyUSB Arduino (in arduino-pico 6.3.1), device and MIDI sources.
S-14	USB Device Class Definition for MIDI Devices, Release 1.0 (1999-11-01). https://www.usb.org/sites/default/files/midi10.pdf
S-16	Raspberry Pi Pico 2 datasheet RP-008299-DS-3 (local copy).
S-17	Raspberry Pi RP2040 datasheet RP-008371-DS-1 (local copy).
EE-S08	Raspberry Pi Pico pinout. https://pico.pinout.xyz/
EE-S14b	Yageo leaded resistors datasheet (MFR series).
EE-S17	Vishay Beyschlag MBA/MBB/MBE professional thin-film resistors, doc 28766. https://www.vishay.com/docs/28766/mbxsma.pdf
EE-S17b	Same document, local copy (sha256 201a7ea8…9cff in the register); 1.2/50 µs pulse chart read from the vector paths: 0414 4.08 kV, 0207 1.74 kV, 0204 1.47 kV at 3.3 kΩ; 0414 minimum lead pitch 15.0 mm.
EE-S24	Vishay BCcomponents LSR37 high-voltage surge resistor, doc 28735 (third-party mirror, local copy): 12 kV ESD contact discharge rating, ±10 %/±20 %.
EE-S25	IEC 60664-1 Table F.2, clearance against rated impulse voltage. Values recalled, standard not opened: to be checked.
EE-S26	Distributor listings for sibling values of the MBE 0414 (RS, LCSC), basis of the 0.30–0.66 EUR per-part price; 3k3 price assumed equal.